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On a Four-Valued Logic of Formal Inconsistency and Formal Undeterminedness

Abstract. Belnap–Dunn’s relevance logic, BD, was designed seeking a suitable logical device for dealing with multiple information sources which sometimes may provide inconsistent and/or incomplete pieces of information. BD is a four-valued logic which is both paraconsistent and paracomplete. On the other hand, De and Omori, while investigating what classical negation amounts to in a paracomplete and paraconsistent four-valued setting, proposed the expansion BD2 of the four valued Belnap–Dunn logic by a classical negation. In this paper, we introduce a four-valued expansion of BD called $\text{BD}^\odot$, obtained by adding an unary connective $\odot$ which is a consistency operator (in the sense of the Logics of Formal Inconsistency, **LFIs**). In addition, this operator is the unique one with the following features: it extends to BD the consistency operator of **LF11**, a well-known three-valued **LFI**, still satisfying axiom **ciw** (which states that any sentence is either consistent or contradictory), and allowing to define an undeterminedness operator (in the sense of Logic of Formal Undeterminedness, **LFUs**). Moreover, $\text{BD}^\odot$ is maximal w.r.t. **LF11**, and it is proved to be equivalent to BD2, up to signature. After presenting a natural Hilbert-style characterization of $\text{BD}^\odot$ obtained by means of twist-structures semantics, we propose a first-order version of $\text{BD}^\odot$ called $\text{QBD}^\odot$, with semantics based on an appropriate notion of four-valued Tarskian-like structures called **4**-structures. We show that in $\text{QBD}^\odot$, the existential and universal quantifiers are interdefinable in terms of the paracomplete and paraconsistent negation, and not by means of the classical negation. Finally, a Hilbert-style calculus for $\text{QBD}^\odot$ is presented, proving the corresponding soundness and completeness theorems.

Keywords: Belnap–Dunn logic, Logics of formal inconsistency, Logics of formal undeterminedness, Twist structures, First-order logic.

1. Introduction

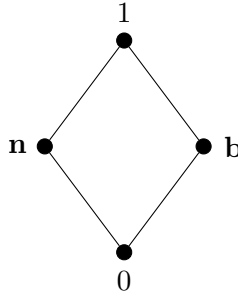
In this paper, we are interested in an specific expansion of the four-valued logic known as Belnap–Dunn logic (BD hereafter). BD, also known as *FDE* (*First Degree Entailment*), was originally developed by Dunn, and later stepped further by Belnap to apply this logic to computer science. This

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is a logical system that is well-known for its many applications in different fields such as the development of languages allowing self-reference, semantics of logic programming and, mainly, it is a basic tool in the area of intelligent database management or question-answering systems. Databases, especially large ones, have a great propensity to become inconsistent and/or incomplete: first, the information stored is usually obtained from different sources which might conflict with each other; second, the information obtained from each source, even if it is not obviously inconsistent, may hide contradictions.

In order to deal with this situation, Belnap proposed his logic on four non-classical epistemic truth-values: 1 (true and not false); 0 (false and not true), where these two values are to some extent identifiable with the classical ones; **n** (neither true nor false), the well-known “undetermined” value of some three-valued logics; and **b** (both true and false) also called “overdetermined”, the value corresponding to the situation where several (probably independent) sources assign a different classical value to a sentence.

Let us denote by **4** the set of truth values $\{1, \mathbf{b}, \mathbf{n}, 0\}$ and consider the signature $\{\vee, \wedge, \neg\}$. In a sense, we can say that the lattice $\mathcal{BD} = \langle \mathbf{4}, \{\vee, \wedge, \neg\} \rangle$ given by



and where $\neg 0 = 1$, $\neg 1 = 0$, $\neg \mathbf{b} = \mathbf{b}$ and $\neg \mathbf{n} = \mathbf{n}$, is an algebraic counterpart of BD. As it is well-known, we can consider two kinds of orders, namely the *truth order* and the *information order*, which is an interesting aspect of BD, and therefore, we can think of the four-element Belnap lattice as a bilattice.

The designated values are 1 (‘just true’) and **b** (‘both true and false’). In turn, the values 0 and **n** are to be taken as ‘just false’ and ‘neither true nor false’, respectively. Thus, when we speak of a sentence being true, we mean that it takes either the value 1 or **b**, and when we speak of a sentence being

false, we mean that it takes either the value **b** or 0. Because of this, $\neg$ is a paraconsistent and paracomplete negation (see Definition 2.9 below).¹

An interesting expansion of BD was proposed in [27] by Font and Rius, namely the *Tetravalent Modal Logic* ($\mathcal{TM}\mathcal{L}$ for short). In their work, these authors devoted to the study of the logic that preserves degrees of truth w.r.t. the class of tetravalent modal algebras (introduced by A. Monteiro and studied by others). The resulting logic turns out to be an expansion of BD by means of a modal operator $\Box$. $\mathcal{TM}\mathcal{L}$ was later studied by Coniglio and M. Figallo in [17, 25] under the perspective of paraconsistent logics. Recently, in [26] it was proposed a cut-free sequent calculus for $\mathcal{TM}\mathcal{L}$ as well as a natural deduction system for it.

On the other hand, De and Omori investigated in [20] the notion of classical negation from a non-classical perspective. In particular, they aimed to determine what classical negation amounts to in a paracomplete and paraconsistent four-valued setting. They considered different negations for BD being each of them “classical” in some respect. These authors conclude that the Boolean complementation (on the lattice $\mathcal{BD}$) is the only negation in this four-valued setting that fulfills all conditions they think a classical negation should verify. Finally, they gave a general semantic characterization of classical negation considering different expansions of four valued Belnap–Dunn logic by classical negation. However, one of the negations considered in [20], denoted $\neg^2$, is the one defined by $\neg^2 1 = 0 = \neg^2 \mathbf{b}$, $\neg^2 \mathbf{n} = \mathbf{b}$ and $\neg^2 0 = 1$, and then BD2 is defined as the expansion of BD by this negation.

In this paper, we introduce a different four-valued expansion of BD called $\mathbf{BD}^\odot$, obtained by adding an unary connective $\odot$ which is a consistency operator (in the sense of the Logics of Formal Inconsistency, **LFIs**). In addition, it will be shown that this operator is the unique one with the following features: it extends to BD the consistency operator $\circ$ of **LF1**, a well-known three-valued **LFI**, still satisfying axiom **ciw** (which states that any sentence is either consistent or contradictory), and allowing to define an undeterminedness operator (in the sense of Logic of Formal Undeterminedness, **LFUs**).

This paper is organized as follows. In Section 2 we state all the notions, definitions and basic results about **LFIs** and **LFUs** that will be used along the paper. In Section 3, we introduce the logic $\mathbf{BD}^\odot$, an expansion of BD in terms of the consistency operator $\odot$ and prove in Remark 3.15 that this operator is the unique one satisfying the above-mentioned properties.

¹The possibility of considering another set of designated values over this structure was considered by several authors. A good reference on this topic is [39]. We thank one of the anonymous referees for drawing our attention to this reference.

Besides, we prove that $\text{BD}^\odot$ coincides with BD2 up to language, that it is maximal w.r.t LFI1, and also provide two *Derivability Adjustment Theorems* (DATS) of $\text{BD}^\odot$ w.r.t CPL and LFI1, respectively. In Section 4, we develop a twist-structure-style semantics for $\text{BD}^\odot$ and apply it in order to obtain a sound and complete Hilbert-style calculus for $\text{BD}^\odot$. In Sects. 5 and 6, we extend $\text{BD}^\odot$ to first-order languages. That is, we present the first-order logic $\text{QBD}^\odot$, with semantics based on an appropriate notion of Tarskian-like structures based on the four-valued algebra underlying $\text{BD}^\odot$, called **4**-structures. Then, we show that, in $\text{QBD}^\odot$, $\exists$ and $\forall$ are interdefinable in terms of the paracomplete and paraconsistent negation, and not by means of the classical negation. Finally, a Hilbert-style calculus for $\text{QBD}^\odot$ is presented, proving the corresponding soundness and completeness theorems.

2. Preliminaries

Let $\mathcal{L}$ be a propositional language over a given signature Θ and let $\text{For}(\Theta)$ (or simply For) be the set of all well-formed formulas (or simply formulas) over $\mathcal{L}$. In this paper, a logic $\mathbf{L}$ is a pair $\mathbf{L} = \langle \text{For}, \vdash_{\mathbf{L}} \rangle$ where $\vdash_{\mathbf{L}}$ is a subset of $2^{\text{For}} \times \text{For}$. In what follows, lowercase Greek letters will stand for formulas and uppercase Greek letters will stand for sets of formulas. As usual, we shall write $\Gamma, \psi \vdash_{\mathbf{L}} \varphi$ instead of $\Gamma \cup \{\psi\} \vdash_{\mathbf{L}} \varphi$; $\psi_1, \dots, \psi_n \vdash_{\mathbf{L}} \varphi$ instead of $\{\psi_1, \dots, \psi_n\} \vdash_{\mathbf{L}} \varphi$; and $\vdash_{\mathbf{L}} \varphi$ instead of $\emptyset \vdash_{\mathbf{L}} \varphi$.

Recall that a logic $\mathbf{L}$ is *Tarskian* if it satisfies: (1) if $\alpha \in \Gamma$, then $\Gamma \vdash_{\mathbf{L}} \alpha$; (2) if $\Gamma \vdash_{\mathbf{L}} \alpha$ and $\Gamma \subseteq \Delta$, then $\Delta \vdash_{\mathbf{L}} \alpha$; and (3) if $\Delta \vdash_{\mathbf{L}} \alpha$ and $\Gamma \vdash_{\mathbf{L}} \beta$ for all $\beta \in \Delta$, then $\Gamma \vdash_{\mathbf{L}} \alpha$. $\mathbf{L}$ is *structural* if $\text{For} = \text{For}(\Theta)$ for a signature Θ and the following holds: if $\Gamma \vdash_{\mathbf{L}} \varphi$ then $\sigma[\Gamma] \vdash_{\mathbf{L}} \sigma(\varphi)$ for every substitution σ over Θ .² A logic is *standard* if it is Tarskian and structural. Finally, a logic $\mathbf{L}$ is *compact* (or *finitary*) if it holds: if $\Gamma \vdash_{\mathbf{L}} \alpha$, there is a finite set $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0 \vdash_{\mathbf{L}} \alpha$.

Given a logic $\mathbf{L} = \langle \text{For}, \vdash_{\mathbf{L}} \rangle$, a *theory* of $\mathbf{L}$ is any subset of For . A theory Γ is *closed* if: $\Gamma \vdash_{\mathbf{L}} \alpha$ iff $\alpha \in \Gamma$. A theory Γ is φ -*saturated* in $\mathbf{L}$ if $\Gamma \not\vdash_{\mathbf{L}} \varphi$, but $\Gamma, \alpha \vdash_{\mathbf{L}} \varphi$ for every α such that $\alpha \notin \Gamma$. It is easy to prove that any φ -saturated theory in a Tarskian logic is closed. The following useful result will be used along this paper. A proof can be found, for instance, in [41, Theorem 22.2]):

²A substitution over Θ is a function $\sigma : \text{For}(\Theta) \rightarrow \text{For}(\Theta)$ such that $\sigma(\#(\psi_1, \dots, \psi_n)) = \#(\sigma(\psi_1), \dots, \sigma(\psi_n))$, for every n -ary connective $\# \in \Theta$ and formulas $\psi_1, \dots, \psi_n$.

THEOREM 2.1. (Lindenbaum-Łoś) *Let $\mathbf{L}$ be a Tarskian and finitary logic, and let $\Gamma \cup \{\varphi\}$ be a set of formulas such that $\Gamma \not\vdash_{\mathbf{L}} \varphi$. Then, there exists a set of formulas Δ such that Δ is φ -saturated in $\mathbf{L}$ and $\Gamma \subseteq \Delta$.*

Let $\mathbb{P}$ be a propositional signature. An *algebra* for $\mathbb{P}$ is a pair $\mathbf{A} = \langle A, (\cdot)^{\mathbf{A}} \rangle$ where (1) A is a nonempty set; and (2) $(\cdot)^{\mathbf{A}}$ is a function which maps every n -ary connective of $\mathbb{P}$ into an n -ary operation on A , i.e. if $c \in \mathbb{P}$ is an n -ary connective, then $c^{\mathbf{A}}$ is a function $c^{\mathbf{A}} : A^n \rightarrow A$.

As usual, as long as there is no room for confusion, we shall use the same symbol to design a connective c and its interpretation $c^{\mathbf{A}}$. Because of this, from now on an algebra for $\mathbb{P}$ will be denoted by $\mathbf{A} = \langle A, \mathbb{P} \rangle$.

EXAMPLE 2.2. Consider the propositional signature $\mathbb{P}_2 = \{\wedge, \vee, \neg\}$. Let **DM4** be the four-element De Morgan algebra given by $\mathbf{DM4} = \langle \mathbf{4}, \mathbb{P}_2 \rangle$ where the operations are defined in Table 1. This algebra underlies Belnap-Dunn's logic BD, to be formally introduced in Definition 2.15 below.

A *logic matrix* (or simply a *matrix*) over the propositional signature $\mathbb{P}$ is a pair $\mathcal{M} = \langle \mathbf{A}, \mathcal{D} \rangle$ where $\mathbf{A} = \langle \mathcal{V}, (\cdot)^{\mathcal{M}} \rangle$ is an algebra for $\mathbb{P}$ and $\mathcal{D} \subseteq \mathcal{V}$. The domain $\mathcal{V}$ of the algebra is called *the set of truth values* and the elements of $\mathcal{D}$ are the *designated values* of the matrix. When there is no doubt about the signature we are working with, the logical matrix will be simply denoted by $\mathcal{M} = \langle \mathcal{V}, \mathcal{D} \rangle$.

Let $\mathcal{M} = \langle \mathcal{V}, \mathcal{D} \rangle$ be a matrix over $\mathbb{P}$. An $\mathcal{M}$ -*valuation* (or $\mathcal{M}$ -*morphism*) for $\mathbb{P}$ is a map $v : \text{For}(\mathbb{P}) \rightarrow \mathcal{V}$ such that for every n -ary connective $\# \in \mathbb{P}$ it holds:

$$v(\#(\psi_1, \dots, \psi_n)) = \#(v(\psi_1), \dots, v(\psi_n)).$$

We denote by $\text{Val}(\mathcal{M})$ the set of all $\mathcal{M}$ -valuations. Let $\psi \in \text{For}(\mathbb{P})$, $\Gamma \subseteq \text{For}(\mathbb{P})$ and v an $\mathcal{M}$ -valuation. We say that v is an $\mathcal{M}$ -*model* of ψ (or that v $\mathcal{M}$ -satisfies ψ) if $v(\psi) \in \mathcal{D}$. We denote by $\text{mod}_{\mathcal{M}}(\psi)$ the set of all $\mathcal{M}$ -models of ψ ; and we say that ψ is $\mathcal{M}$ -*satisfiable* if $\text{mod}_{\mathcal{M}}(\psi) \neq \emptyset$. Besides, v is an $\mathcal{M}$ -*model* of Γ if $v \in \text{mod}_{\mathcal{M}}(\alpha)$ for every $\alpha \in \Gamma$. We denote

Table 1. Operations of **DM4**

$\wedge$	1	b	n	0
1	1	b	n	0
b	b	b	0	0
n	n	0	n	0
0	0	0	0	0

$\vee$	1	b	n	0
1	1	1	1	1
b	1	b	1	b
n	1	1	n	n
0	1	b	n	0

	$\neg$
1	0
b	b
n	n
0	1

by $\text{mod}_{\mathcal{M}}(\Gamma) := \bigcap_{\psi \in \Gamma} \text{mod}_{\mathcal{M}}(\psi)$ the set of $\mathcal{M}$ -models of Γ ; and Γ is $\mathcal{M}$ -satisfiable if $\text{mod}_{\mathcal{M}}(\Gamma) \neq \emptyset$.

Every matrix $\mathcal{M}$ over a signature $\mathbb{P}$ induces a logic $\mathbf{L} = \langle \text{For}(\mathbb{P}), \models_{\mathbf{L}} \rangle$ where

$$\Gamma \models_{\mathbf{L}} \psi \quad \text{iff} \quad \text{mod}_{\mathcal{M}}(\Gamma) \subseteq \text{mod}_{\mathcal{M}}(\psi).$$

If $\models_{\mathbf{L}} \psi$, i. e. $v(\psi) \in \mathcal{D}$ for every $\mathcal{M}$ -valuation v , we say that ψ is $\mathbf{L}$ -valid or that it is a $\mathbf{L}$ -tautology. If $\mathbf{L}$ is the logic induced by the matrix $\mathcal{M}$, $\mathcal{M}$ -valuations may alternatively be called $\mathbf{L}$ -valuations. The following is a well-known fact (see, for instance, [42, Theorems 3.1.3 and 4.1.7]).

THEOREM 2.3. *Let $\mathcal{M} = \langle \mathcal{V}, \mathcal{D} \rangle$ be a finite matrix over $\mathbb{P}$. Then, the logic $\mathbf{L} = \langle \text{For}(\mathbb{P}), \models_{\mathbf{L}} \rangle$ induced by $\mathcal{M}$ is standard and finitary.*

The notion of maximality between logics has been studied in the literature (see, for instance, [16, Def. 2.7]) and it will be relevant in the present paper.

DEFINITION 2.4. Let $\mathbf{L}_1$ and $\mathbf{L}_2$ be two standard logics defined over the same propositional signature $\mathbb{P}$ such that $\vdash_{\mathbf{L}_1} \subsetneq \vdash_{\mathbf{L}_2}$. Then $\mathbf{L}_1$ is said to be *maximal* w.r.t. $\mathbf{L}_2$ if, for every $\varphi \in \text{For}(\mathbb{P})$ such that $\vdash_{\mathbf{L}_2} \varphi$ but $\not\vdash_{\mathbf{L}_1} \varphi$, it holds that the logic $\mathbf{L}_1^+$, obtained from $\mathbf{L}_1$ adding φ as a theorem, coincides with $\mathbf{L}_2$.

DEFINITION 2.5. Let $\mathbf{L}_1$ and $\mathbf{L}_2$ be two standard logics defined over the same propositional signature $\mathbb{P}$ such that $\vdash_{\mathbf{L}_1} \subsetneq \vdash_{\mathbf{L}_2}$ and let $\textcircled{\mathbf{R}}$ be a (primitive or defined) connective. It is called a Derivability Adjustment Theorem (DAT) to any of the following:

$$\begin{aligned} \Gamma \vdash_{\mathbf{L}_2} \psi & \quad \text{iff} \quad \text{there exists } \Lambda \subseteq \text{For}(\mathbb{P}) \\ & \quad \text{such that } \Gamma, \Lambda^{\textcircled{\mathbf{R}}} \vdash_{\mathbf{L}_1} \psi \text{ where } \Lambda^{\textcircled{\mathbf{R}}} = \{\textcircled{\mathbf{R}}\gamma : \gamma \in \Lambda\}. \end{aligned}$$

$$\Gamma \vdash_{\mathbf{L}_2} \psi \quad \text{iff} \quad \Gamma, \{\textcircled{\mathbf{R}}p_1, \dots, \textcircled{\mathbf{R}}p_m\} \vdash_{\mathbf{L}_1} \psi, \text{ where } p_1, \dots, p_m \text{ are the propositional letters occurring in } \Gamma \cup \{\psi\}.$$

LEMMA 2.6. ([16, Lemma 2.3]) *Let $\mathbf{L}_1$ and $\mathbf{L}_2$ be the logics induced by the matrices $\langle \mathbf{A}_1, \mathcal{D}_1 \rangle$ and $\langle \mathbf{A}_2, \mathcal{D}_2 \rangle$, respectively, defined over the same signature $\mathbb{P}$. If $\mathbf{A}_2$ is a subalgebra of $\mathbf{A}_1$ and $\mathcal{D}_2 = \mathcal{D}_1 \cap \mathbf{A}_2$, then $\models_{\mathbf{L}_1} \subseteq \models_{\mathbf{L}_2}$.*

THEOREM 2.7. ([16, Th. 2.4]) *Let $\mathbf{L}_1$ and $\mathbf{L}_2$ be the logics induced by the matrices $\langle \mathbf{A}_1, \mathcal{D}_1 \rangle$ and $\langle \mathbf{A}_2, \mathcal{D}_2 \rangle$, respectively, defined over the same propositional signature $\mathbb{P}$, such that $\mathbf{A}_2$ is a subalgebra of $\mathbf{A}_1$ and $\mathcal{D}_2 = \mathcal{D}_1 \cap \mathbf{A}_2$. Suppose that the following conditions hold:*

- (1) $A_1 = \{0, 1, a_1, \dots, a_k, a_{k+1}, \dots, a_n\}$ and $A_2 = \{0, 1, a_1, \dots, a_k\}$ are finite, $0 \notin \mathcal{D}_1$, $1 \in \mathcal{D}_2$ and $\{0, 1\}$ is a subalgebra of A_2 , where $0 \leq k < n$ (if $k = 0$ then $A_2 = \{0, 1\}$).
- (2) There are $\top(p), \perp(p) \in \text{For}(\mathbb{P})$ such that $e(\top(p)) = 1$ and $e(\perp(p)) = 0$ for all $\mathbf{L}_1$ -valuation e .
- (3) There exists $\alpha_j^i(p) \in \text{For}(\mathbb{P})$ such that $e(p) = a_i$ only if $e(\alpha_j^i(p)) = a_j$, for $k + 1 \leq i \leq n$, $1 \leq j \leq n$, and $i \neq j$.

Then, $\mathbf{L}_1$ is maximal w.r.t. $\mathbf{L}_2$.

2.1. Logics of Formal Inconsistency and Undeterminedness

It is convenient at this point to recall the class of paraconsistent logics known as *Logics of Formal Inconsistency*, as well as the class of paracomplete logics called *Logics of Formal Undeterminedness*. To this aim, some basic definitions will be stated first.

DEFINITION 2.8. Let $\mathbf{L} = \langle \text{For}, \vdash_{\mathbf{L}} \rangle$ be a propositional Tarskian logic.

- (a) a unary connective $\neg$ (primitive or defined) in the formal language $\mathcal{L}$ is a *negation*,³ if there is a formula α in the language such that $\alpha \not\vdash_{\mathbf{L}} \neg \alpha$ and $\neg \alpha \vdash_{\mathbf{L}} \alpha$.
- (b) a binary connective $\vee$ (primitive or defined) in the formal language $\mathcal{L}$ is a *disjunction* if it holds:

$$\Delta, \alpha \vdash_{\mathbf{L}} \gamma \text{ and } \Delta, \beta \vdash_{\mathbf{L}} \gamma \quad \text{iff} \quad \Delta, \alpha \vee \beta \vdash_{\mathbf{L}} \gamma.$$

DEFINITION 2.9. Let $\mathbf{L} = \langle \mathcal{L}, \vdash_{\mathbf{L}} \rangle$ be a Tarskian logic with negation connective $\neg$ (either primitive or defined) in the formal language $\mathcal{L}$.

- (i) $\neg$ is a *classical negation* if $\mathbf{L}$ has a *disjunction* $\vee$ (either primitive or defined) and it holds⁴

$$\begin{array}{ll} \text{(tnd)} & \vdash_{\mathbf{L}} \alpha \vee \neg \alpha \quad (\text{tertium non datur}) \\ \text{(ecq)} & \alpha, \neg \alpha \vdash_{\mathbf{L}} \beta \quad (\text{ex contradictione sequitur quodlibet}) \end{array}$$

³This definition, taken from [5, Definition 15.1] establishes natural (necessary) conditions for a negation connective, but of course it does not characterizes precisely what a negation should be. The latter question is far beyond the scope of this paper.

⁴It should be observed that the requirement of having a disjunction $\vee$ in order to define a classical negation is due to the fact that we are dealing with single-conclusion consequence relations. In the context of multiple-conclusion consequence relations, this notion can be introduced without using disjunctions.

- (ii) $\neg$ is a *paraconsistent negation* if it does not satisfies the principle **(ecq)**, i.e., $\alpha, \neg\alpha \not\vdash_{\mathbf{L}} \beta$ for some α and β .
- (iii) $\neg$ is a *paracomplete negation* if $\mathbf{L}$ has a *disjunction* $\vee$ (either primitive or defined) and it does not satisfy the principle **(tnd)**, i.e., $\not\vdash_{\mathbf{L}} \beta \vee \neg\beta$ for some β .⁵
- (iv) $\neg$ is a *parafinite negation* if $\mathbf{L}$ has a *disjunction* $\vee$ (either primitive or defined) and $\neg$ is paraconsistent and paracomplete.

The first hierarchy of paraconsistent calculi is the family C_n (for $n \geq 1$), introduced by da Costa in his groundbreaking Cathedral Thesis [18]. In each logic C_n it is possible to define a unary connective $\circ_n$ which recovers, in a local way, the explosiveness of its paraconsistent negation $\neg$, in the following sense: $\alpha, \neg\alpha, \circ_n\alpha \vdash_{C_n} \beta$ for every formulas α, β of C_n . According to da Costa's terminology, $\circ_n\alpha$ denotes that α is *well-behaved* in C_n . This approach to paraconsistency was generalized by Carnielli and Marcos through the notion of *Logics of Formal Inconsistency*.

DEFINITION 2.10. ([12]) Let $\mathbf{L}$ be a finitary Tarskian logic with a paraconsistent negation connective $\neg$. We say that $\mathbf{L}$ is a *Logic of Formal Inconsistency* (**LFI**, for short) if there is a unary connective $\circ$ (either primitive or defined), called *consistency operator*, such that

- (a) $\alpha, \circ\alpha \not\vdash_{\mathbf{L}} \beta$, for some α, β ,
- (b) $\neg\alpha, \circ\alpha \not\vdash_{\mathbf{L}} \beta$, for some α, β ,
- (c) $\alpha, \neg\alpha, \circ\alpha \vdash_{\mathbf{L}} \beta$, for every α, β ,

In [10] it was studied a basic **LFI**, called **mbC**. It is obtained by adding to the standard Hilbert calculus for positive classical propositional logic (see the axioms in Proposition 3.4 below, where *Modus Ponens* its the only inference rule) a negation $\neg$ satisfying **(tnd)** (as an axiom schema) and a consistency operator $\circ$ satisfying the *gentle explosion law*, namely the axiom schema $(\circ\alpha \wedge (\alpha \wedge \neg\alpha)) \rightarrow \beta$. Several axiomatic extensions of **mbC** were studied in the literature (see, for instance, [2, 4, 9, 10, 12] and [3]). Most of these systems cannot be semantically characterized by a single finite logical matrix. However, they are characterized by a suitable 2-valued (non-truth-functional) valuation semantics, as well as by means of diverse (non-deterministic) semantical tools (see, for instance, [9, Chapter 6]).

⁵Once again, it is necessary to have a disjunction to express paracompleteness given that we are dealing with single-conclusion consequence relations. This is no longer required in the context of multiple-conclusion consequence relations.

REMARK 2.11. (The axiom **ciw**) Avron proposes in [2, Definition 2.21], within a framework of multiple-conclusion calculi, two rules that could be added to **mbC**, namely

$$\begin{aligned} (\mathbf{k1}) \quad & \vdash \circ\alpha, \alpha \\ (\mathbf{k2}) \quad & \vdash \circ\alpha, \neg\alpha \end{aligned}$$

Both rules were reintroduced in [4], within a framework of Hilbert calculi, as the axioms

$$\begin{aligned} (\mathbf{k1}) \quad & \circ\alpha \vee \alpha \\ (\mathbf{k2}) \quad & \circ\alpha \vee \neg\alpha \end{aligned}$$

In [3] both axioms were condensed into a single one, called (**k**):

$$(\mathbf{k}) \quad \circ\alpha \vee (\alpha \wedge \neg\alpha)$$

This axiom states that any formula is either consistent or contradictory (and it cannot be both, by definition of **LFI**s). Axiom (**k**) was rebaptized as **ciw** in [9, Chapter 3], where the axiomatic extension of **mbC** by adding **ciw** was called **mbCciw**, and some of its basic (meta)properties were analyzed. In particular, it was shown that it is the least extension of **mbC** in which $\circ\alpha$ is equivalent to $\sim(\alpha \wedge \neg\alpha)$, where $\sim$ is a classical negation definable in **mbC** (and so in its extensions). In a more general setting observe that, if **L** is an **LFI** with a conjunction $\wedge$, then $\circ\alpha$ is a kind of ‘meet-complement’ of the contradiction $\alpha \wedge \neg\alpha$, in the sense that $\circ\alpha \wedge (\alpha \wedge \neg\alpha) \vdash_{\mathbf{L}} \beta$, for every α, β (hence, $\circ\alpha \wedge (\alpha \wedge \neg\alpha)$ is a bottom). The role played by axiom **ciw** is stating that $\circ\alpha$ is also a sort of ‘join-complement’ of $\alpha \wedge \neg\alpha$, by adding the property $\vdash_{\mathbf{L}} \circ\alpha \vee (\alpha \wedge \neg\alpha)$ (i.e., $\circ\alpha \vee (\alpha \wedge \neg\alpha)$ is a top). Hence, in any **LFI** satisfying **ciw**, $\circ\alpha$ is promoted to a kind of ‘Boolean complement’ of $\alpha \wedge \neg\alpha$, in the sense of having the logical behavior of the (classical) negation of the latter formula. Because of this nice feature, **ciw** turns out to be an important axiom in the universe of **LFI**s and **LFIU**s.

EXAMPLE 2.12. (The 3-valued logic **LFI1**, part 1) The three-valued paraconsistent logic **LFI1**, introduced in [13], was independently proposed by various authors. Among others, it is equivalent (up to signature) to the well-known paraconsistent logic **J3** Introduced by D’Ottaviano and da Costa in [21] (for more details about this topic consult, for instance, [10, Example 18] and [9, Sections 4.4.3 and 4.4.7]).

Recall from Example 2.2 the algebra $\mathbf{DM4} = \langle 4, \{\wedge, \vee, \neg\} \rangle$. Observe that $\mathbf{3} := \{1, \mathbf{b}, 0\}$ is a subalgebra of **DM4**. As shown in [9, Section 4.4.7], the

Table 2. Operations of $\mathbf{A}_{\mathbf{LF11}}$

$\rightarrow$	1	b	0
1	1	b	0
b	1	b	0
0	1	1	1

	$\circ$
1	1
b	0
0	1

algebra $\mathbf{A}_{\mathbf{LF11}}$ underlying the characteristic matrix of $\mathbf{LF11}$ can be defined over the signature $\{\wedge, \vee, \rightarrow, \neg, \circ\}$ by expanding $\mathbf{3}$ with the operations $\rightarrow$ and $\circ$ displayed in Table 2. The matrix defining the logic $\mathbf{LF11}$ is $\mathcal{M}_{\mathbf{LF11}} = \langle \mathbf{A}_{\mathbf{LF11}}, \mathcal{D}_{\mathbf{LF11}} \rangle$ where $\mathcal{D}_{\mathbf{LF11}} = \{1, \mathbf{b}\}$.

From the axiomatic perspective, $\mathbf{LF11}$ is obtained from $\mathbf{mbC}$ by adding some axioms, which imply $\mathbf{ciw}$ and the axiom schema $\mathbf{cc}: \circ\circ\alpha$. It is worth noting that the truth-table for $\circ$ in $\mathcal{M}_{\mathbf{LF11}}$ is the only one corresponding to a consistency operator that satisfies simultaneously $\mathbf{ciw}$ and $\mathbf{cc}$. This 3-valued consistency operator is quite natural: it is intuitively clear that 0 and 1, which correspond to the classical truth-values, should be consistent truth-values (that is, $\circ(0)$ and $\circ(1)$ should be designated), which produces the table above.

The dual notion of $\mathbf{LFIs}$ was proposed by Marcos, through the concept of *Logic of Formal Undeterminedness*:

DEFINITION 2.13. ([31]) Let $\mathbf{L}$ be a finitary Tarskian logic with a disjunction connective $\vee$ and a paracomplete negation connective $\neg$. We say that $\mathbf{L}$ is a *Logic of Formal Undeterminedness* ($\mathbf{LFU}$, for short) if there is a unary connective $\star$ (either primitive or defined), called *undeterminedness operator*,⁶ such that

- (d) $\not\vdash_{\mathbf{L}} \alpha \vee \star\alpha$, for some α ,
- (e) $\not\vdash_{\mathbf{L}} \neg\alpha \vee \star\alpha$, for some α ,
- (f) $\vdash_{\mathbf{L}} \alpha \vee \neg\alpha \vee \star\alpha$, for every α .

DEFINITION 2.14. ([11]) Let $\mathbf{L}$ be a finitary Tarskian logic with a disjunction connective $\vee$ and a parafinite negation connective $\neg$. We say that $\mathbf{L}$ is a *Logic of Formal Inconsistency and Undeterminedness* ($\mathbf{LFIU}$, for short) if

⁶Marcos uses in [31] the terminology *determinedness* for this kind of operator, while defining an undeterminedness operator as what we call here a determinedness operator. The terminology we adopted here was proposed in [11], and was also adopted in [15]. The idea is that, if both α and $\neg\alpha$ are false, then it is the case that α is undetermined, and so $\star\alpha$ is true.

there are unary connectives $\circ$ and $\star$ (either primitive or defined), such that **L** is an **LFI** w.r.t. $\neg$ and $\circ$, and an **LFU** w.r.t. $\neg$ and $\star$.

In the same way as consistency and undeterminedness operators $\circ$ and $\star$ were introduced, it could be possible to consider their (classically) negated operators $\bullet$ and $\star$ of *inconsistency* and *determinedness*, which are characterized by the following properties:

- (a') $\alpha \not\vdash_{\mathbf{L}} \bullet\alpha$, for some α ,
- (b') $\neg\alpha \not\vdash_{\mathbf{L}} \bullet\alpha$, for some α ,
- (c') $\alpha, \neg\alpha \vdash_{\mathbf{L}} \bullet\alpha$, for every α ,
- (d') $\star\alpha \not\vdash_{\mathbf{L}} \alpha$, for some α ,
- (e') $\star\alpha \not\vdash_{\mathbf{L}} \neg\alpha$, for some α ,
- (f') $\star\alpha \vdash_{\mathbf{L}} \alpha \vee \neg\alpha$, for every α .

In the **LFIUs** studied in this paper it will hold that $\bullet\alpha$ and $\star\alpha$ can be defined, respectively, by $\neg\circ\alpha$ and $\neg\star\alpha$.

According to what Arieli et al. stated in [1, Theorem 1], there are necessarily at least four truth values for a logic to enjoy the properties of paraconsistency and paracompleteness, simultaneously. The paradigm of these four-valued logics, denoted **BD**, was initially developed by Dunn (cf. [22, 23]) as the semantic for the *first degree entailment* logic, being this applied in the realm of Computer Science by Belnap in his seminal papers (cf. [6, 7]).

DEFINITION 2.15. (Belnap-Dunn's 4-valued logic **BD**, or *FDE*) The logic **BD** is the pair $\langle \text{For}(\mathbb{P}_2), \models_{\mathbf{BD}} \rangle$ where the consequence relation $\models_{\mathbf{BD}}$ is determined by the matrix $\mathcal{M}_{\mathbf{BD}} = \langle \mathbf{DM4}, \mathcal{D}_{\mathbf{BD}} \rangle$. Here, **DM4** is the algebra of Example 2.2 and the set of designated values is $\mathcal{D}_{\mathbf{BD}} = \{1, \mathbf{b}\}$.

THEOREM 2.16. *The primitive connective $\neg$ of **BD** is a parafinite negation.*

PROOF. Let p, q two propositional letters and let v be a **BD**-valuation such that $v(p) = \mathbf{b}$ and $v(q) = \mathbf{n}$, then $v \in \text{mod}_{\mathbf{BD}}(p \wedge \neg p)$. However, $v \notin \text{mod}_{\mathbf{BD}}(q)$; and therefore $\neg$ does not satisfy **(ecq)**. On the other hand, if we choose a propositional letter p and a valuation v such that $v(p) = \mathbf{n}$, it is clear that $v \notin \text{mod}_{\mathbf{BD}}(p \vee \neg p)$ and hence $\neg$ rejects **(tnd)**. ■

REMARK 2.17. (The logic **BS4**) In [35], Omori and Waragai introduced the four-valued logic **BS4** pursuing to modify the three-valued logic **LFI1** (see Example 2.12) in such a way that it can deal with information that not only

is inconsistent but also incomplete. In terms of matrices, this modification corresponds to a move from a three-valued matrix to a four valued-matrix. In this way, they introduced a four-valued matrix whose induced logic is an expansion of **BD** but in the language of **LFI1**, where the operation $\circ$ is defined in such a way that the matrix of **LFI1**, $\mathcal{M}_{\text{LFI1}}$, is a sub-matrix of the new one. Moreover, axiom $\circ\circ\alpha$ still holds, and so $\circ$ can only take values 0 or 1.

In precise terms, let $\mathbb{P}_3$ be the propositional signature $\mathbb{P}_3 = \{\wedge, \vee, \hookrightarrow, \neg, \circ, -\}$, and consider the algebra $\mathbf{A}_{\text{BS4}} = \langle \mathbf{4}, \mathbb{P}_3 \rangle$ whose reduct $\langle \mathbf{4}, \{\wedge, \vee, \neg\} \rangle$ is the algebra **DM4** and the operations $\hookrightarrow$, $-$ and $\circ$ are given in Table 3. Let $\mathcal{M}_{\text{BS4}} = \langle \mathbf{A}_{\text{BS4}}, \mathcal{D}_{\text{BS4}} \rangle$ be the matrix whose set of designated values is $D_{\text{BS4}} = \{1, \mathbf{b}\}$.

Then, $\text{BS4} = \langle \text{For}(\mathbb{P}_3), \models_{\text{BS4}} \rangle$ is the logic induced by the matrix $\mathcal{M}_{\text{BS4}}$ (see [35, Definition 5]). Observe that **ciw** is no longer valid in **BS4**, hence this logic is not a deductive extension of **LFI1**.

The simplicity and naturalness of **BS4** is evidenced by the fact that this same logic was discovered almost simultaneously by different authors from different points of view and in diverse contexts. For instance, in his dissertation (see [19]) da Silva Oliveira presents the logic **BD_~** that turns out to be equivalent to **BS4**, but his motivations are in the context of logic programming. On the other hand, in [20] De and Omori present **BS4** under the name of **BDe** as one possible extension of **BD** by means of some kind of classical negation. Other systems equivalent to **BS4** that can be found in the literature are: **BIDM4** (see [36, p. 70]), **B₄[→]** (see [34]) and **BD Δ** (see [38]).

As we shall see in the next two theorems, **BS4** is an **LFIU** in which the consistency and determinedness operators coincide, as well as undeterminedness and inconsistency do. Indeed, let $\bullet\alpha := \neg\circ\alpha$. Then:

THEOREM 2.18. *BS4 is an LFIU where $\circ$ and $\bullet$ are the respective operators of consistency and undeterminedness. Moreover, $\bullet$ is also an inconsistency operator.*

Table 3. Operations of $\mathbf{A}_{\text{BS4}}$

$\hookrightarrow$	1	b	n	0
1	1	b	n	0
b	1	b	n	0
n	1	1	1	1
0	1	1	1	1

	$\circ$	$-$
1	1	0
b	0	0
n	0	1
0	1	1

PROOF. Since the underlying algebra of the matrix of BD is a reduct of the underlying algebra of the matrix of BS4, we have that the negation $\neg$ is parafinite in BS4. Let p and q be two propositional letters and let e_1 and e_2 two BS4-valuations such that $e_1(p) = 1$, $e_1(q) = 0$, $e_2(p) = 0$ and $e_2(q) = 0$. We can see that $\text{mod}_{\text{BS4}}(\{p, \circ p\}) \not\subseteq \text{mod}_{\text{BS4}}(q)$, $\text{mod}_{\text{BS4}}(\{\neg p, \circ p\}) \not\subseteq \text{mod}_{\text{BS4}}(q)$, $e_2 \not\models \text{mod}_{\text{BS4}}(\{p \vee \bullet p\})$, $e_1 \not\models \text{mod}_{\text{BS4}}(\{\neg p \vee \bullet p\})$. On the other hand $\text{mod}_{\text{BS4}}(\{p, \neg p, \circ p\}) = \emptyset$ and $\text{mod}_{\text{BS4}}(p \vee \neg p \vee \bullet p) = \text{Val}(\mathcal{M}_{\text{BS4}})$. Therefore, all conditions of Definition 2.14 are fulfilled. It is immediate to prove that $\bullet$ is also an inconsistency operator. ■

THEOREM 2.19. *In BS4, the operator $\circ$ recovers (ecq) and (tnd). That is, $\circ$ is also a determinedness operator.*

PROOF. In every LFI that extends classical logic it holds that the consistency operator recovers (ecq). Suppose that $v(\circ\alpha) = 1$, then $v(\alpha) \in \{0, 1\}$. Therefore, $\circ\alpha \vdash_{\text{BS4}} \alpha \vee \neg\alpha$. It is immediate to see that $\circ p \vdash_{\text{BS4}} p$ and $\circ p \not\vdash_{\text{BS4}} \neg p$, for every propositional letter p : it is enough to consider the BS4-valuations e_1 and e_2 such that $e_1(p) = 0$ and $e_2(p) = 1$. Hence, $\circ$ is a determinedness operator. ■

REMARK 2.20. In a series of papers, Carnielli and Rodrigues introduced a family of logics designed to deal with paraconsistency and paracompleteness from an epistemic and informational perspective (see, for instance, [14]). These logics are called *Logics of Evidence and Truth* (LETs for short). LETs are expansions of BD by adding a (primitive or not) *classicality operator* $\circ$ which simultaneously recovers (ecq) and (tnd). In that sense, the logic BS4 is a LET. As we shall see in Remark 3.9, the logic BD° to be considered in the following sections is also a LET.

3. The Propositional Logic BD°

The logic BD2 was introduced in [20] as an expansion of BD by means of a classical negation which satisfies certain requirements. In turn, in the present paper we were looking for an expansion of BD (in the context of the LFIs) by means of a consistency operator $\circ$ that, unlike what happens with $\circ$ in BS4, it can take values other than 0 or 1. In this specific sense, $\circ$ can be seen as a *weak* consistency operator (see Remark 3.2). In addition, we require that $\circ$ satisfies interesting properties in the context of LFIUs. This being so, we arrive to a system called BD° that, as it will be shown in Remark 3.3, is equivalent to BD2, but obtained by following different motivations. This equivalence is due to the fact that the classical negation proposed in [20]

and our consistency operator are inter-definable. Consequently, we can assert that BD2 is an **LFU** with an undeterminedness operator definable from a consistency operator. The interesting properties enjoyed by $\odot$ in the realm of **LFU**s will be summarized in Remark 3.15.

Let us consider the propositional signature $\mathbb{P}_T = \{\vee, \wedge, \neg, \odot\}$.

DEFINITION 3.1. The logic $\text{BD}^\odot = \langle \text{For}(\mathbb{P}_T), \models_{\text{BD}^\odot} \rangle$ is the logic induced by the matrix $\mathcal{M}_{\text{BD}^\odot} = \langle \mathbf{A}_{\text{BD}^\odot}, \mathcal{D}_{\text{BD}^\odot} \rangle$ where $\mathbf{A}_{\text{BD}^\odot} = \langle \mathbf{4}, \mathbb{P}_T \rangle$ is the algebra whose reduct $\langle \mathbf{4}, \{\wedge, \vee, \neg\} \rangle$ is **DM4**, the operation $\odot$ is given in Table 4 and the set of designated values is $\mathcal{D}_{\text{BD}^\odot} = \{1, \mathbf{b}\}$.

Consider now the operators $\sim, \circ, \star, \star$ and $\rightarrow$ (see Table 5).

It is immediate to see that they are definable in $\mathbf{A}_{\text{BD}^\odot}$ as follows:

$$\begin{aligned} \sim x &:= \odot x \wedge \neg(x \wedge \odot x), & \star x &:= \odot \odot x, \\ \star x &:= \neg \odot \odot x, & \circ x &:= \odot x \wedge \star x, \\ x \rightarrow y &:= \sim x \vee y. \end{aligned}$$

REMARK 3.2. From the truth-tables of $\text{BD}^\odot$ it is immediate to see that $\frac{}{\text{BD}^\odot} \circ \circ \alpha$ but $\not\frac{}{\text{BD}^\odot} \odot \odot \alpha$ and $\not\frac{}{\text{BD}^\odot} \circ \odot \alpha$. In this sense, it can be said that $\odot$ is a weak consistency operator.

REMARK 3.3. ($\text{BD}^\odot$ is De-Omori's BD2, up to language) In [20, p. 833] De and Omori propose four possible classical-like negations that can be added to BD. This produces four different expansions of BD with a classical negation, in the sense of Definition 2.9(i). It is worth noting that the negation $\sim$, defined above in $\text{BD}^\odot$, is one of the four classical-like negations proposed

Table 4. $\odot$ operator of $\mathbf{A}_{\text{BD}^\odot}$

	$\odot$
1	1
b	0
n	b
0	1

Table 5. Some operations definable in $\mathbf{A}_{\text{BD}^\odot}$

	$\sim$	$\circ$	$\star$	$\star$
1	0	1	0	1
b	0	0	0	1
n	b	0	1	0
0	1	1	0	1

$\rightarrow$	1	b	n	0
1	1	b	n	0
b	1	b	n	0
n	1	b	1	b
0	1	1	1	1

on page 833 of that paper, that they call $\neg^2$. Moreover, on page 843 the authors observe that the logic BD2, which is obtained by expanding BD with the classical-like negation $\neg^2$ (i.e., $\sim$), is a proper extension of BS4, since $\sim$ is not definable in $\mathbf{A}_{BS4}$. Given that $\sim$ is definable in $\mathbf{A}_{BD^\odot}$, it follows that $BD^\odot$ is an extension of BD2. In turn, $\odot$ can be defined in BD2 by $\odot x := \sim(x \wedge \neg x)$. This shows that $BD^\odot$ and BD2 are the same logic, up to signature, which properly extends BS4. This means that all the results to be proven in the sequel for $BD^\odot$ are also valid for BD2.

PROPOSITION 3.4. *$BD^\odot$ is sound w.r.t the positive classical propositional logic, i.e. in $BD^\odot$ the following axiom schemata are valid*

Pos1	$\alpha \rightarrow (\beta \rightarrow \alpha)$	Pos7	$\alpha \rightarrow (\beta \vee \alpha)$
Pos2	$(\alpha \rightarrow (\beta \rightarrow \gamma)) \rightarrow$ $((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \gamma))$	Pos8	$(\alpha \rightarrow \gamma) \rightarrow ((\beta \rightarrow \gamma) \rightarrow$ $(\alpha \vee \beta \rightarrow \gamma))$
Pos3	$\alpha \wedge \beta \rightarrow \alpha$	Pos9	$\alpha \vee (\alpha \rightarrow \beta)$
Pos4	$\alpha \wedge \beta \rightarrow \beta$		
Pos5	$\alpha \rightarrow (\beta \rightarrow (\alpha \wedge \beta))$		
Pos6	$\alpha \rightarrow (\alpha \vee \beta)$		

PROOF. It is a straightforward verification. ■

As mentioned in Remark 3.3, $\sim$ has features of a classical negation. This will be formally proven in the following result (see also Remark 3.6).

PROPOSITION 3.5. *The connective $\sim$ is a classical negation for $BD^\odot$.*

PROOF. It is clear, from the definition of $\sim$, that $\text{mod}_{BD^\odot}(\{\alpha, \sim\alpha\}) = \emptyset$ and $\text{mod}_{BD^\odot}(\alpha \vee \sim\alpha) = \text{Val}(\mathcal{M}_{BD^\odot})$, for all $\alpha \in \text{For}(\mathbb{P}_T)$. Therefore, $\sim$ satisfies **(ecq)** and **(tnd)**. ■

REMARK 3.6. In $BD^\odot$, the negation $\sim$ is a classical one in the sense that it validates *ex contradictione sequitur quodlibet* $\alpha, \sim\alpha \vdash_{BD^\odot} \beta$ and *tertium non datur* $\vdash_{BD^\odot} \alpha \vee \sim\alpha$.

PROPOSITION 3.7. *The logic $BD^\odot$ is an **LFIU** in which $\odot$ is a consistency operator satisfying axiom **ciw** $\odot\alpha \vee (\alpha \wedge \neg\alpha)$, and $\star$ is the undeterminedness operator. Moreover, $\star x := \neg\star x$ is a determinedness operator, and $\neg\odot$ is an inconsistency operator.*

PROOF. It is immediate to see that $\neg\odot$ is an inconsistency operator. The negation $\neg$ is paracomplete since it coincides with the negation of BD. Let p and q be two propositional letters and consider the $BD^\odot$ -valuation v such that $v(p) = 1$ and $v(q) = 0$. Then, it is clear that $\text{mod}_{BD^\odot}(\{\odot p, p\}) \not\subseteq$

$\text{mod}_{\text{BD}^\odot}(q)$ and $v \notin \text{mod}_{\text{BD}^\odot}(\neg p \vee \star p)$ (or equivalently, $\text{mod}_{\text{BD}^\odot}(\star p) \not\subseteq \text{mod}_{\text{BD}^\odot}(\neg p)$). On the other hand, by considering the $\text{BD}^\odot$ -valuation w such that $w(p) = 0$ and $w(q) = 1$ we can see that $\text{mod}_{\text{BD}^\odot}(\{\odot p, \neg p\}) \not\subseteq \text{mod}_{\text{BD}^\odot}(q)$ and $w \notin \text{mod}_{\text{BD}^\odot}(p \vee \star p)$ (or equivalently, $\text{mod}_{\text{BD}^\odot}(\star p) \not\subseteq \text{mod}_{\text{BD}^\odot}(p)$).

Finally, since for all $\alpha \in \text{For}(\mathbb{P}_T)$ we have $\text{mod}_{\text{BD}^\odot}(\{\alpha, \neg\alpha, \odot \neg\alpha\}) = \emptyset$, $\text{mod}_{\text{BD}^\odot}(\alpha \vee \neg\alpha \vee \star \alpha) = \text{Val}(\mathcal{M}_{\text{BD}^\odot})$ (or $\text{mod}_{\text{BD}^\odot}(\star \alpha) \subseteq \text{mod}_{\text{BD}^\odot}(\alpha \vee \neg\alpha)$) and $\text{mod}_{\text{BD}^\odot}(\odot \alpha \vee (\alpha \wedge \neg\alpha)) = \text{Val}(\mathcal{M}_{\text{BD}^\odot})$, the assertion holds. ■

PROPOSITION 3.8. $\text{BD}^\odot$ is finitary.

PROOF. From Theorem 2.3. ■

REMARK 3.9. By Theorem 2.19 adapted to $\text{BD}^\odot$, the operator $\circ$ is a classicality operator in the sense that it recovers (**ecq**) and (**tnd**). This means that $\text{BD}^\odot$ is a **LET**, recall Remark 2.20. The relationship between $\text{BD}^\odot$ and others **LETs** studied in the literature is a subject that deserves future research.

EXAMPLE 3.10. (The 3-valued logic **LF11**, part 2) Let us return to the logic **LF11** introduced in Example 2.12. Recall from Definition 3.1 that $\mathbf{A}_{\text{BD}^\odot}$ is the algebra over signature $\mathbb{P}_T$ underlying the matrix logic $\text{BD}^\odot$. Observe that $\mathbf{3} = \{1, \mathbf{b}, 0\}$ is a subalgebra of $\mathbf{A}_{\text{BD}^\odot}$ in which $\odot x = \circ x$, for every $x \in \mathbf{3}$. From this, **LF11** can be defined over signature $\mathbb{P}_T$. Indeed, given that the implication $\rightarrow$ of $\text{BD}^\odot$ is definable in $\mathbf{A}_{\text{BD}^\odot}$ over $\mathbb{P}_T$, its restriction to $\mathbf{3}$, which coincides with the primitive implication of **LF11**, is also definable in the implication-less reduct of $\mathbf{A}_{\text{LF11}}$ (an algebra over $\mathbb{P}_T$). The corresponding matrix logic with domain $\mathbf{3}$ and set of designated values $D_{\text{BD}^\odot} = \{1, \mathbf{b}\}$ is therefore a presentation of **LF11** over $\mathbb{P}_T$ as a sub-matrix of $\text{BD}^\odot$. As we shall see in Remark 3.15, $\text{BD}^\odot$ is the only **LF1U** extending **LF11** to the four-valued domain $\mathbf{4}$ where its consistency operator satisfies **ciw**. From now on, **LF11** will be considered as a matrix logic over $\mathbb{P}_T$ as described above.

The next proposition is a first step in order to prove that $\text{BD}^\odot$ is maximal with respect **LF11** by using Theorem 2.7.

PROPOSITION 3.11. *CPL and LF11 are proper deductive extensions of $\text{BD}^\odot$ over $\mathbb{P}_T$.*

PROOF. Clearly, $\langle \{1, 0\}, \mathbb{P}_T \rangle$ is a sub-algebra of $\mathbf{A}_{\text{BD}^\odot} = \langle \mathbf{4}, \mathbb{P}_T \rangle$. Besides, $\mathcal{D}_{\text{CPL}} = \{1\} = \mathcal{D}_{\text{BD}^\odot} \cap \{0, 1\}$. By Lemma 2.6, we have that **CPL** is a deductive extension of $\text{BD}^\odot$. The proof that **LF11** is a deductive extension of $\text{BD}^\odot$ is analogous. Clearly, $\text{BD}^\odot$ is properly contained in both logics under the given interpretation of the connectives: given two different propositional letters p

and q , then $(p \wedge \neg p) \rightarrow q$ holds in CPL but it does not hold in $\text{BD}^{\odot}$. In turn, $\odot \odot p$ holds in LFI1 but it does not hold in $\text{BD}^{\odot}$. ■

Moreover, the following interesting property holds:

THEOREM 3.12. $\text{BD}^{\odot}$ is maximal w.r.t. LFI1.

PROOF. Let us see that the conditions of Theorem 2.7 are verified. In the proof of Proposition 3.11, we checked that the basic conditions for each respective matrix hold. Besides, it satisfies conditions (1)-(3) of Theorem 2.7. Indeed: (1) $\mathbf{A}_{\text{BD}^{\odot}} = \{1, 0, \mathbf{b}, \mathbf{n}\}$ and $\mathbf{A}_{\text{LFI1}} = \{1, 0, \mathbf{b}\}$ are finite, $0 \notin \mathcal{D}_{\text{BD}^{\odot}}$, $1 \in \mathcal{D}_{\text{LFI1}}$ and $\{0, 1\}$ is a subalgebra of $\mathbf{A}_{\text{LFI1}}$; (2) Let $\top(p) := \star p \vee \star p$ and $\perp(p) := p \wedge \neg p \wedge \odot p$, then $v(\top(p)) = 1$ and $v(\perp(p)) = 0$ for all $\text{BD}^{\odot}$ -valuation v ; (3) Let us consider the formula $\varphi_{\mathbf{b}}^{\mathbf{n}}(p) = \odot p$. Then, for every $\text{BD}^{\odot}$ -valuation v , $v(p) = \mathbf{n}$ implies that $v(\varphi_{\mathbf{b}}^{\mathbf{n}}(p)) = \mathbf{b}$. Therefore, $\text{BD}^{\odot}$ is maximal w.r.t. LFI1 by Theorem 2.7. ■

As observed in Example 3.10, $\mathbf{3} = \{1, \mathbf{b}, 0\}$ is a subalgebra of $\mathbf{A}_{\text{BD}^{\odot}}$ such that $\odot x = \circ x$. Hence, the matrix of LFI1 is a sub-matrix of $\text{BD}^{\odot}$, a four-valued **LFIU**. Notice that if a consistency operator $\circ$ defined in an expansion of **DM4** is such that $\circ(\mathbf{n})$ is not designated, then axiom **ciw** does not hold. Hence, it is immediate to see that there are two possible ways to extend the 3-valued operator $\circ$ of LFI1 to **4**, in order to obtain an **LFI** with a **ciw**-consistency operator expanding BD, namely by means of one the following operators:

	$\odot$	$\odot_1$
1	1	1
$\mathbf{b}$	0	0
$\mathbf{n}$	$\mathbf{b}$	1
0	1	1

As shown above, $\odot$ allows to define $\star$, an undeterminedness operator in $\text{BD}^{\odot}$. It is immediate to see that, according to Definition 2.13, the only possibilities for expanding BD by an undeterminedness operator $\otimes$ are the following:

	$\otimes$
1	$0/\mathbf{n}$
$\mathbf{b}$	*
$\mathbf{n}$	$1/\mathbf{b}$
0	$0/\mathbf{n}$

where x/y means that either x or y are admissible values, and $*$ means that any value can be considered.

It can be seen that $\odot$ is more general (or more expressive) than $\odot_1$ in the following sense: while it is possible to define an undeterminedness operator (namely, $\star$) in the expansion of **BD** by $\odot$, it is not possible to define any undeterminedness operator $\otimes$ in the expansion of **BD** by $\odot_1$. Indeed:

LEMMA 3.13. *Let $\mathbf{A}_{\mathbf{DM4}}^1$ be the expansion of the algebra **DM4** by $\odot_1$. Then, for every term $t(x)$ over the signature of $\mathbf{A}_{\mathbf{DM4}}^1$ the following holds:*

- (1) *If $t(\mathbf{n}) = \mathbf{n}$ then $t(0) \in \{0, 1\}$;*
- (2) *If $t(\mathbf{n}) = 1$ then $t(0) = 1$;*
- (3) *If $t(\mathbf{n}) = 0$ then $t(0) = 0$.*

PROOF. Observe first that $\{0, \mathbf{n}, 1\}$ is a subalgebra of $\mathbf{A}_{\mathbf{DM4}}^1$. Then, $t(\mathbf{n}) \neq \mathbf{b}$ for every term $t(x)$. Now, let $t(x)$ be a term. By induction on the complexity n of $t(x)$ it will be shown that conditions (1)-(3) are satisfied.

Base case $n = 0$: hence, $t(x) = x$ (a propositional variable). Then, conditions (1)-(3) hold trivially.

Let $t(x)$ be a term with complexity $n+1$ for $n \geq 0$. We analyze the following cases:

Case $t(x) = \neg s(x)$. If $t(\mathbf{n}) = \mathbf{n}$ then $s(\mathbf{n}) = \mathbf{n}$ and so, by induction hypothesis, $s(0) \in \{0, 1\}$. Therefore $t(0) \in \{0, 1\}$. Now, if $t(\mathbf{n}) = 1$ then $s(\mathbf{n}) = 0$ and so, by induction hypothesis, $s(0) = 0$. From this, $t(0) = 1$. The case $t(\mathbf{n}) = 0$ is proved analogously.

Case $t(x) = \odot_1 s(x)$. Observe that $t(\mathbf{n}) \neq \mathbf{n}$, by definition of $\odot_1$. If $t(\mathbf{n}) = 1$ then $s(\mathbf{n}) \in \{0, \mathbf{n}, 1\}$. If $s(\mathbf{n}) = \mathbf{n}$ then, by induction hypothesis, $s(0) \in \{0, 1\}$ and then $t(0) = 1 \in \{0, 1\}$. If $s(\mathbf{n}) = 0$ then, by induction hypothesis, $s(0) = 0$ and so $t(0) = 1 \in \{0, 1\}$. If $s(\mathbf{n}) = 1$ then, by induction hypothesis, $s(0) = 1$ and then $t(0) = 1 \in \{0, 1\}$. This shows condition (2). Now, if $t(\mathbf{n}) = 0$ then $s(\mathbf{n}) = \mathbf{b}$. But this is impossible, since $\{0, \mathbf{n}, 1\}$ is a subalgebra of $\mathbf{A}_{\mathbf{DM4}}^1$.

Case $t(x) = s(x) \wedge r(x)$. If $t(\mathbf{n}) = \mathbf{n}$ then $s(\mathbf{n}), r(\mathbf{n}) \in \{1, \mathbf{n}\}$, by definition of $\wedge$ in **BD**. By induction hypothesis, $s(0), r(0) \in \{0, 1\}$ and therefore $t(0) \in \{0, 1\}$. If $t(\mathbf{n}) = 1$ then $s(\mathbf{n}) = 1 = r(\mathbf{n})$, by definition of $\wedge$ in **BD**. By

induction hypothesis, $s(0) = 1 = r(0)$ and therefore $t(0) = 1$. Now, if $t(\mathbf{n}) = 0$ then $s(\mathbf{n}) = 0$ or $r(\mathbf{n}) = 0$, by definition of $\wedge$ in **BD**. By induction hypothesis, either $s(0) = 0$ or $r(0) = 0$ and then $t(0) = 0$.

Case $t(x) = s(x) \vee r(x)$. It is proved analogously. ■

PROPOSITION 3.14. *It is not possible to define an undeterminedness operator $\otimes$ as above in the algebra $\mathbf{A}_{\mathbf{DM4}}^1$.*

PROOF. It is a consequence of Lemma 3.13 and the requirements for $\otimes$ discussed above, taking into account that $t(\mathbf{n}) \neq \mathbf{b}$ for every term $t(x)$ over the signature of $\mathbf{A}_{\mathbf{DM4}}^1$. ■

In Remark 3.2 it was observed that, in $\mathbf{BD}^\odot$, the consistency operator $\circ$ satisfies axiom **cc**: $\circ\circ\alpha$, but $\odot$ does not, since $\odot\odot\alpha$ is not a valid schema. In turn, $\odot$ validates axiom schema **ciw** while $\circ$ does not. On the other hand, $\odot_1$ is an interesting consistency operator in $\mathbf{BD}^\odot$ since it satisfies both axioms, **ciw** and **cc**. Hence, by Proposition 3.1.10(3) in [9], $\odot_1$ also satisfies **ci**: $\neg\odot_1\alpha \rightarrow (\alpha \wedge \neg\alpha)$, an axiom schema valid in **LFI1**.

REMARK 3.15. Proposition 3.14 shows that $\odot$ is the only operator that satisfies the following:

- (1) it allows to define an expansion $\mathbf{BD}^\odot$ of **BD** which is maximally contained in **LFI1**;
- (2) it extends the three-valued operator $\circ$ of **LFI1** to **4**, being a consistency operator in $\mathbf{BD}^\odot$ satisfying axiom **ciw**; and
- (3) it allows to define an undeterminedness operator in $\mathbf{BD}^\odot$.

These features justify the relevance of the connective $\odot$.

The logic $\mathbf{BD}^\odot$ satisfies the following Derivability Adjustment Theorems (recall Definition 2.5):

THEOREM 3.16. *If Γ is a finite set of formulas over $\mathbb{P}_T$, then*

$$\Gamma \frac{}{\mathbf{LFI1}} \psi \text{ iff } \Gamma, \star p_1, \dots, \star p_k \frac{}{\mathbf{BD}^\odot} \psi$$

where $p_1, \dots, p_k$ are the propositional variables occurring in $\Gamma \cup \{\psi\}$.

PROOF. “Only if” part. Let v a $\mathcal{M}_{\mathbf{BD}^\odot}$ -valuation such that $v[\Gamma \cup \{\star p_1, \dots, \star p_k\}] \subseteq \{1, \mathbf{b}\}$. Then $v(p_i) \in \{1, \mathbf{b}, 0\}$ for all $1 \leq i \leq n$. Let $\bar{v}$ be the $\mathcal{M}_{\mathbf{LFI1}}$ -valuation defined by

$$\bar{v}(p) := \begin{cases} v(p) & \text{if } p \in \{p_1, \dots, p_k\} \\ 0 & \text{if } p \notin \{p_1, \dots, p_k\} \end{cases}$$

It is clear that $\bar{v}(\alpha) = v(\alpha)$ for every formula α such that $\text{var}(\alpha) \in \{p_1, \dots, p_k\}$. Then, $\bar{v}[\Gamma] \subseteq \{1, \mathbf{b}\}$ and by hypothesis we have $\bar{v}(\psi) \in \{1, \mathbf{b}\}$. Therefore, $v(\psi) \in \{1, \mathbf{b}\}$.

“If” part. Let v be a $\mathcal{M}_{\text{LFII}}$ -valuation such that $v[\Gamma] \subseteq \{1, \mathbf{b}\}$. Let us consider the $\mathcal{M}_{\text{BD}^\odot}$ -valuation $\bar{v}$ where $\bar{v} = v$. Then, $\bar{v}[\Gamma] \subseteq \{1, \mathbf{b}\}$ and $\bar{v}(\star p_i) = 1$ for each $1 \leq i \leq k$. By hypothesis, it follows that $v(\psi) = \bar{v}(\psi) \in \{1, \mathbf{b}\}$. ■

THEOREM 3.17. *Consider the logic CPL presented over the signature $\mathbb{P}_T$ with matrix $\mathcal{M}_{\text{CPL}} = \langle \mathbf{2}, \mathcal{D}_{\text{CPL}} \rangle$, where $\mathbf{2} = \langle \{0, 1\}, \mathbb{P}_T \rangle$ is the subalgebra of $\mathbf{A}_{\text{BD}^\odot}$. If Γ is a finite set of formulas, then*

$$\Gamma \Vdash_{\text{CPL}} \psi \text{ iff } \Gamma, \circ p_1, \dots, \circ p_k \Vdash_{\text{BD}^\odot} \psi$$

where $p_1, \dots, p_k$ are the propositional variables occurring in $\Gamma \cup \{\psi\}$.

PROOF. It is similar to the proof of Theorem 3.16. ■

Obviously Theorem 3.16 can be generalized to any (finite or non-finite) set of formulas:

THEOREM 3.18. *Let $\Gamma \cup \{\psi\}$ be a set of formulas. Then,*

$$\Gamma \Vdash_{\text{LFII}} \psi \text{ iff } \Gamma, (\text{var}(\Gamma))^\star, (\text{var}(\psi))^\star \Vdash_{\text{BD}^\odot} \psi$$

where $(\text{var}(\Gamma))^\star = \{\star p : p \in \text{var}(\Gamma)\}$ and $(\text{var}(\psi))^\star = \{\star p : p \in \text{var}(\psi)\}$. Here, $\text{var}(\Gamma)$ and $\text{var}(\psi)$ are the sets of propositional variables occurring in Γ and ψ , respectively.

The same generalization can be made with Theorem 3.17.

4. Twist Structures and a Hilbert Calculus for $\text{BD}^\odot$

In this section a Hilbert style presentation for $\text{BD}^\odot$ will be given, obtained from a semantical characterization by means of twist structures.

Twist structures were introduced independently by Fidel ([24]) and Vakarelov ([40]) in 1977–78 with the aim of characterizing Nelson’s logic **N4**. However, the main algebraic ideas underlying twist structures were already introduced in 1958 by Kalman ([29]). Indeed, in such a work Kalman introduced the now called De Morgan lattices starting from a distributive lattice L in which he defined the following operations over $L \times L$:

- (1) $\smile(a, b) = (b, a)$;
- (2) $(a, b) \tilde{\wedge} (c, d) = (a \sqcap c, b \sqcup d)$;
- (3) $(a, b) \tilde{\vee} (c, d) = (a \sqcup c, b \sqcap d)$

where $\sqcap$ and $\sqcup$ denote infima and suprema in L . This produces a De Morgan lattice. The same idea was proposed by Fidel and Vakarelov for $\mathbf{N4}$, and also by Dunn in [22], where he obtained a representation of De Morgan lattices by means of pairs of sets with the same kind of operations. Twist structures semantics were afterwards proposed for several logics. In the sequel, a twist structure semantics for $\mathbf{BD}^{\odot}$ will be introduced.

First, consider for convenience the signature $\mathbb{P}_T^{\rightarrow} = \{\vee, \wedge, \rightarrow, \neg, \odot\}$ for $\mathbf{BD}^{\odot}$. The presentation of $\mathbf{BD}^{\odot}$ as a matrix logic over signature $\mathbb{P}_T^{\rightarrow}$ will be denoted by $\mathbf{BD}^{\odot}_T$. The idea is, as in the case of $\mathbf{BD}$, identifying the truth values of $\mathbf{4}$ with pairs (a, b) in $\mathbf{2} = \{0, 1\}$ such that a represents (in informal terms) information for (or in favor of) a given sentence α while b represents information about the negation of α (or against α). Hence, 1 , $\mathbf{b}$, $\mathbf{n}$ and 0 are identified, respectively with $(1, 0)$, $(1, 1)$, $(0, 0)$ and $(0, 1)$.⁷ Consider the following twist operators over $\mathbf{2} \times \mathbf{2}$: $\tilde{\neg}$, $\tilde{\wedge}$, $\tilde{\vee}$ are defined by means of (1)-(3) above, and

$$(4) \quad \odot(a, b) = (\sim(a \sqcap b), a \Leftrightarrow b)$$

Note that, by using these operations and the definitions of $\sim$ and $\rightarrow$ given in Section 3,

$$(5) \quad \tilde{\neg}(a, b) = (\sim a, b \Rightarrow a) \text{ and}$$

$$(6) \quad (a, b) \tilde{\rightarrow} (c, d) = (a \Rightarrow c, (b \Rightarrow a) \sqcap d)$$

where $a \Rightarrow b = \sim a \sqcup b$ and $\sim a$ denote respectively the implication and the Boolean complement in $\mathbf{2}$ (seen as a Boolean algebra), and $a \Leftrightarrow b = (a \Rightarrow b) \sqcap (b \Rightarrow a)$. Let $\mathcal{T}_2$ be the (twist) algebra over the signature $\mathbb{P}_T^{\rightarrow}$ obtained in this way, and let $\mathcal{M}_2 = \langle \mathcal{T}_2, D_2 \rangle$ be the logical matrix associated to $\mathcal{T}_2$ where $D_2 = \{(1, 0), (1, 1)\}$. Observe that, with the identifications above, these operations correspond to the truth-tables of the operators in the signature $\mathbb{P}_T^{\rightarrow}$ given in Section 3, hence $\mathcal{M}_2$ is (up to names) the logical matrix for $\mathbf{BD}^{\odot}_T$.

This construction can be easily defined over any Boolean algebra $\mathbf{A}$: indeed, let $\mathcal{T}_{\mathbf{A}} = \langle A \times A, \tilde{\vee}, \tilde{\wedge}, \tilde{\rightarrow}, \tilde{\neg}, \odot \rangle$ be the (twist) algebra over the signature $\mathbb{P}_T^{\rightarrow}$ such that the operations are defined by means of clauses (1)-(4) and (6) above (where now $a, b, c, d \in A$, the domain of $\mathbf{A}$). The logical matrix associated to $\mathcal{T}_{\mathbf{A}}$ is $\mathcal{M}_{\mathbf{A}} = \langle \mathcal{T}_{\mathbf{A}}, D_{\mathbf{A}} \rangle$ such that $D_{\mathbf{A}} = \{(1, a) : a \in A\}$ and 1 is the top element of $\mathbf{A}$. Let $\frac{T}{\mathbf{A}}$ be the consequence relation associated to $\mathcal{M}_{\mathbf{A}}$, namely: $\Gamma \frac{T}{\mathbf{A}} \varphi$ iff, for every valuation v over $\mathcal{M}_{\mathbf{A}}$, if $v(\gamma) \in D_{\mathbf{A}}$

⁷This is analogous to the interpretation of Belnap and Dunn's logic *FDE*.

for every $\gamma \in \Gamma$ then $v(\varphi) \in D_{\mathbf{A}}$. Let $\frac{T}{\text{BD}^{\odot}}$ be the consequence relation associated to the class of twist structures for $\text{BD}^{\odot}_T$, namely: $\Gamma \frac{T}{\text{BD}^{\odot}} \varphi$ iff $\Gamma \frac{T}{\mathbf{A}} \varphi$ for every $\mathbf{A}$.

A Hilbert calculus for $\text{BD}^{\odot}$ (to be precise, for $\text{BD}^{\odot}_T$) can be naturally defined from this semantics. By Proposition 3.4, the logic $\text{BD}^{\odot}_T$ is a model of positive classical logic. Taking this into account, and the definition of the twist operators $\odot$ and $\leadsto$, we obtain the following Hilbert calculus $\text{HBD}^{\odot}_T$ for $\text{BD}^{\odot}_T$ (where $\alpha \leftrightarrow \beta$ is an abbreviation for $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$):

Axioms

Axiom schemata **Pos1-Pos9** from positive classical logic (recall Proposition 3.4), plus the following ones:

(DNeg)	$\neg\neg\alpha \leftrightarrow \alpha$
(DM1)	$\neg(\alpha \vee \beta) \leftrightarrow (\neg\alpha \wedge \neg\beta)$
(DM2)	$\neg(\alpha \wedge \beta) \leftrightarrow (\neg\alpha \vee \neg\beta)$
(DM3)	$\neg(\alpha \rightarrow \beta) \leftrightarrow ((\neg\alpha \rightarrow \alpha) \wedge \neg\beta)$
($\odot$ - 1)	$(\odot \alpha \wedge (\alpha \wedge \neg\alpha)) \rightarrow \beta$
(ciw)	$\odot \alpha \vee (\alpha \wedge \neg\alpha)$
($\odot$ - 2)	$\neg\odot \alpha \leftrightarrow (\alpha \leftrightarrow \neg\alpha)$

Rule of inference (MP) $\frac{\alpha \quad \alpha \rightarrow \beta}{\beta}$ (*Modus Ponens*)

REMARK 4.1. Observe that the six axiom schemata above are naturally induced by the twist operations, by taking into account that the coordinates of each ordered pair in 2×2 represent the value of α and $\neg\alpha$, respectively. The first four axiom schemata are inherited from BD . In turn, ($\odot$ - 1) and (ciw) logically characterize the formula $\odot \alpha$, represented by the first component of $\odot(a, b)$, provided that a and b correspond to α and $\neg\alpha$, respectively. According to (4), this formula must be equivalent to the Boolean negation of $(\alpha \wedge \neg\alpha)$. Finally, ($\odot$ - 2) logically characterizes the formula $\neg\odot \alpha$, represented by the second component of $\odot(a, b)$, assuming that a and b represent, respectively, α and $\neg\alpha$. By (4), this value must correspond to the formula $(\alpha \leftrightarrow \neg\alpha)$.

In order to prove soundness and completeness of $\text{HBD}^{\odot}_T$ w.r.t. the four-valued matrix logic $\text{BD}^{\odot}_T$ it will be first proven soundness and completeness of $\text{HBD}^{\odot}_T$ w.r.t. the twist structures semantics defined above. Hence, the former result will be obtained from the latter by using standard arguments

from Boolean algebras.⁸ Before doing this, a useful notation will be introduced. Let $v : For(\mathbb{P}_T^{\rightarrow}) \rightarrow \mathcal{T}_{\mathbf{A}}$ be a valuation over $\mathcal{M}_{\mathbf{A}}$ (that is, a homomorphism of $\mathbb{P}_T^{\rightarrow}$ -algebras). From now on, v can be denoted by $v = (v_1, v_2)$ where $v_1, v_2 : For(\mathbb{P}_T^{\rightarrow}) \rightarrow A$. Hence, $v(\alpha) = (v_1(\alpha), v_2(\alpha))$ for every formula α .

THEOREM 4.2. (Soundness and completeness of $\text{HBD}^{\odot}_T$ w.r.t. twist structures semantics) *Let $\Gamma \cup \{\varphi\}$ be a set of formulas over $\mathbb{P}_T^{\rightarrow}$. Then: $\Gamma \vdash_{\text{HBD}^{\odot}_T} \varphi$ iff $\Gamma \vdash_{\text{BD}^{\odot}}^T \varphi$.*

PROOF. (*Soundness*): Let $\mathbf{A}$ be a Boolean algebra. It is immediate to see that, for any instance ψ of an axiom of $\text{HBD}^{\odot}_T$, it is the case that $v_1(\psi) = 1$ for every valuation v over $\mathcal{M}_{\mathbf{A}}$. That is, $v(\psi) \in D_{\mathbf{A}}$ for every axiom ψ . For instance, since $v(\odot \alpha) = \odot(v(\alpha))$ and $v(\neg \alpha) = \neg(v(\alpha))$ then $v_1(\odot \alpha) = \sim(v_1(\alpha) \sqcap v_2(\alpha)) = \sim(v_1(\alpha) \sqcap v_1(\neg \alpha))$ (by definition of $\odot$ and $\neg$). Hence, $v_1(\psi) = 1$ for every instance ψ of axioms $(\odot\text{-}1)$ and **(ciw)**. Axioms **Pos1-Pos9** from positive classical logic are clearly valid in every $\mathcal{M}_{\mathbf{A}}$, by virtue of the definition of the operations in the twist structures and by the fact that every Boolean algebra is a model of such axioms. In addition, it is clear that $v_1(\alpha \rightarrow \beta) = v_1(\alpha) = 1$ implies that $v_1(\beta) = 1$, given that $v_1(\alpha \rightarrow \beta) = v_1(\alpha) \Rightarrow v_1(\beta)$. From this, by induction on the length of a derivation in $\text{HBD}^{\odot}_T$ of φ from Γ it can be proven the following: $\Gamma \vdash_{\text{HBD}^{\odot}_T} \varphi$ implies that $\Gamma \vdash_{\mathbf{A}}^T \varphi$ for every $\mathbf{A}$.

(*Completeness*): Assume that $\Gamma \not\vdash_{\text{HBD}^{\odot}_T} \varphi$. By Theorem 2.1 there exists a φ -saturated set Δ in $\text{HBD}^{\odot}_T$ containing Γ . Consider now the following relation in $For(\mathbb{P}_T^{\rightarrow})$: $\alpha \equiv_{\Delta} \beta$ iff $\Delta \vdash_{\text{HBD}^{\odot}_T} \alpha \leftrightarrow \beta$. By using the axioms of positive classical logic it is immediate to see that $\equiv_{\Delta}$ is a congruence over $For(\mathbb{P}_T^{\rightarrow})$ w.r.t. the connectives $\wedge$, $\vee$ and $\rightarrow$. That is, $[\alpha] \sqcap [\beta] = [\alpha \wedge \beta]$, $[\alpha] \sqcup [\beta] = [\alpha \vee \beta]$ and $[\alpha] \Rightarrow [\beta] = [\alpha \rightarrow \beta]$ are well-defined operations, where $[\gamma]$ denotes the equivalence class of the formula γ w.r.t. $\equiv_{\Delta}$. Moreover, $A_{\Delta} := For(\mathbb{P}_T^{\rightarrow}) / \equiv_{\Delta}$ is the domain of a Boolean algebra $\mathbf{A}_{\Delta}$ in which $1 = [\alpha \rightarrow \alpha]$, $0 = [\odot \alpha \wedge \alpha \wedge \neg \alpha]$ and $\sim[\alpha] = [\alpha] \Rightarrow 0$ is the Boolean complement, for any formula α . Define now the twist structure $\mathcal{T}_{\mathbf{A}_{\Delta}}$ over $\mathbf{A}_{\Delta}$, as well as its associated logical matrix $\mathcal{M}_{\mathbf{A}_{\Delta}}$.

By the very definitions, $[\alpha] = 1$ iff $\Delta \vdash_{\text{HBD}^{\odot}_T} \alpha$, iff $\alpha \in \Delta$. Now, it is easy to prove that the function $v_{\Delta} : For(\mathbb{P}_T^{\rightarrow}) \rightarrow A_{\Delta} \times A_{\Delta}$ given by $v_{\Delta}(\alpha) = ([\alpha], [\neg \alpha])$ is a valuation over $\mathcal{M}_{\mathbf{A}_{\Delta}}$. This is a consequence of the

⁸A similar technique was used in [8, Section 5].

axioms of $\text{HBD}^{\odot}_T$ and the definition of the operations in the Boolean algebra $\mathbf{A}_{\Delta}$ (in particular, by axioms $(\odot\text{-1})$, (ciw) and $(\odot\text{-2})$, $[\odot \alpha] = \sim([\alpha] \sqcap [\neg \alpha])$ and $[\neg \odot \alpha] = [\alpha] \Leftrightarrow [\neg \alpha]$; and by definition of $\Rightarrow$ and axiom (DM3) , $[\alpha \rightarrow \beta] = [\alpha] \Rightarrow [\beta]$ and $[\neg(\alpha \rightarrow \beta)] = ([\neg \alpha] \Rightarrow [\alpha]) \sqcap [\neg \beta]$). From this, v_{Δ} is a valuation over $\mathcal{M}_{\mathbf{A}_{\Delta}}$ such that $v_{\Delta}(\alpha) \in D_{\mathbf{A}_{\Delta}}^T$ for every $\alpha \in \Gamma$, but $v_{\Delta}(\varphi) \notin D_{\mathbf{A}_{\Delta}}$, since $\varphi \notin \Delta$. This proves that $\Gamma \not\models_{\text{BD}^{\odot}} \varphi$. ■

Now, it will be convenient to recall some classical results concerning Boolean algebras. The interested reader can consult, for instance, [28]:

PROPOSITION 4.3. *Let $\mathbf{A}$ be a Boolean algebra with domain A . Then:*

- (1) *If a is an element of A different from 1, there exists an ultrafilter F over $\mathbf{A}$ such that $a \notin F$.*
- (2) *If F is an ultrafilter over $\mathbf{A}$, the characteristic map $h_F : A \rightarrow \{0, 1\}$ of F , given by $h_F(x) = 1$ iff $x \in F$, is a homomorphism of Boolean algebras between $\mathbf{A}$ and the two-element Boolean algebra $\mathbf{2}$.*

THEOREM 4.4. (Soundness and completeness of $\text{HBD}^{\odot}_T$ w.r.t. $\text{BD}^{\odot}_T$) *Let $\Gamma \cup \{\varphi\}$ be a set of formulas in $\text{For}(\mathbb{P}_T^{\rightarrow})$. Then, $\Gamma \models_{\text{HBD}^{\odot}_T} \varphi$ iff $\Gamma \models_{\text{BD}^{\odot}_T} \varphi$.*

PROOF. (*Soundness*): It follows from Theorem 4.2 (Soundness) and from the fact that the four-valued logical matrix $\text{BD}^{\odot}_T$ coincides (up to names) with $\mathcal{M}_2$, the logical matrix associated to the Boolean algebra $\mathbf{2}$.

(*Completeness*): Assume that $\Gamma \not\models_{\text{HBD}^{\odot}_T} \varphi$. By using completeness of $\text{HBD}^{\odot}_T$ w.r.t. twist structures semantics (see Theorem 4.2), it follows that $\Gamma \not\models_{\text{BD}^{\odot}_T} \varphi$. This means that there exists a Boolean algebra $\mathbf{A}$ and a valuation v over $\mathcal{M}_{\mathbf{A}}$ such that $v(\gamma) \in D_{\mathbf{A}}$ for every $\gamma \in \Gamma$, but $v(\varphi) \notin D_{\mathbf{A}}$. That is, $v_1(\gamma) = 1$ for every $\gamma \in \Gamma$, but $v_1(\varphi) \neq 1$. By Proposition 4.3 item (1), there exists an ultrafilter F over $\mathbf{A}$ such that $v_1(\varphi) \notin F$. Let $h_F : A \rightarrow \{0, 1\}$ be the characteristic map of F . By Proposition 4.3 item (2), h_F is a homomorphism of Boolean algebras between $\mathbf{A}$ and the two-element Boolean algebra $\mathbf{2}$. Define now the function $\bar{v} : \text{For}(\mathbb{P}_T^{\rightarrow}) \rightarrow \mathbf{2} \times \mathbf{2}$ given by $\bar{v}(\alpha) = (h_F(v_1(\alpha)), h_F(v_2(\alpha)))$. Taking into account that h_F is a homomorphism of Boolean algebras, that v is a valuation over $\mathcal{M}_{\mathbf{A}}$, and by the definition of the operations in the twist structure $\mathcal{T}_2$, it is easy to prove that $\bar{v}$ is a valuation over the matrix $\mathcal{M}_2$ (that is, over the four-valued matrix of $\text{BD}^{\odot}_T$, up to names). Moreover, the valuation $\bar{v}$ is such that $\bar{v}(\gamma) \in D_2$ for every $\gamma \in \Gamma$ (since $v_1(\gamma) = 1$, hence $h_F(v_1(\gamma)) = 1$), but $\bar{v}(\varphi) \notin D_2$ (since $v_1(\varphi) \notin F$, hence $h_F(v_1(\varphi)) = 0$). From this, $\Gamma \not\models_{\mathcal{M}_2} \varphi$. That is, $\Gamma \not\models_{\text{BD}^{\odot}_T} \varphi$. ■

Thanks to the completeness theorem, the following result (to be used in the next section) can be proven easily:

PROPOSITION 4.5. *The following schemata are provable in $\text{HBD}^{\odot}_T$.*

- (i) $(\odot \alpha \rightarrow \alpha) \rightarrow \alpha$
- (ii) $((\neg \alpha \rightarrow \alpha) \wedge (\neg \odot \alpha \rightarrow \alpha)) \rightarrow \alpha$
- (iii) $(\alpha \wedge \odot \alpha \wedge \neg \odot \alpha) \rightarrow \beta$
- (iv) $(\neg \odot \alpha \rightarrow \odot \alpha) \rightarrow \odot \alpha$
- (v) $(\odot \alpha \rightarrow \neg \alpha) \rightarrow \neg \alpha$
- (vi) $\odot \alpha \leftrightarrow \odot \neg \alpha$
- (vii) $\neg \odot \alpha \leftrightarrow \neg \odot \neg \alpha$
- (viii) $(\alpha \rightarrow \odot \odot \alpha) \wedge (\neg \alpha \rightarrow \odot \odot \alpha)$
- (xi) $(\alpha \wedge \neg \alpha) \leftrightarrow (\neg \odot \alpha \wedge \odot \odot \alpha)$
- (x) $\neg \alpha \rightarrow \neg(\alpha \wedge \beta)$
- (xi) $(\neg \alpha \wedge \odot \alpha) \rightarrow \odot(\alpha \wedge \beta)$
- (xii) $(\odot \alpha \wedge \neg \odot \alpha) \rightarrow \odot(\alpha \wedge \beta)$
- (xiii) $(\odot \alpha \wedge \odot \beta) \rightarrow \odot(\alpha \wedge \beta)$
- (xiv) $((\alpha \wedge \odot \alpha) \wedge \neg \odot \beta) \rightarrow \neg \odot(\alpha \wedge \beta)$
- (xv) $((\odot \alpha \wedge \neg \odot \alpha) \wedge (\odot \beta \wedge \neg \odot \beta)) \rightarrow \neg \odot(\alpha \wedge \beta)$

PROOF. It is easy to prove that each schema (i)-(xv) is a tautology in $\text{BD}^{\odot}_T$. Hence, they are provable in $\text{HBD}^{\odot}_T$, by Theorem 4.4. ■

5. The First-Order Logic $\text{QBD}^{\odot}$

In this section we introduce a quantified version of $\text{BD}^{\odot}$, that will be called $\text{QBD}^{\odot}$. Rigorously speaking, $\text{QBD}^{\odot}$ is a quantified version of $\text{BD}^{\odot}_T$: indeed, by convenience, the (defined) implication $\rightarrow$ will be considered as a primitive one in $\text{QBD}^{\odot}$. However, to lighten up notation the subscript T will be omitted for $\text{QBD}^{\odot}$.

To start with, consider the language $\mathcal{L}_{\mathcal{S}} = \langle \mathcal{Q}, \mathbb{P}_{\vec{T}}, \mathcal{S} \rangle$ where $\mathcal{Q} = \{\forall, \exists\}$ is the set of quantifiers symbols, $\mathbb{P}_{\vec{T}} = \{\vee, \wedge, \rightarrow, \neg, \odot\}$ is the propositional signature of $\text{BD}^{\odot}_T$ and $\mathcal{S} = \mathcal{P} \cup \mathcal{F} \cup \mathcal{C}$ is a first-order signature formed by the (disjoint) sets $\mathcal{P} \neq \emptyset$, $\mathcal{F}$ and $\mathcal{C}$ containing predicate, function and constant symbols, respectively. Besides, let $\mathcal{V}$ be a denumerable set of variable symbols. We denote by $\text{For}(\mathcal{S})$ the set of all (well-formed) formulas of $\mathcal{L}_{\mathcal{S}}$,

$\text{Sent}(\mathcal{S})$ the set of all closed formulas (or sentences) of $\mathcal{L}_{\mathcal{S}}$, $\text{Ter}(\mathcal{S})$ the set of all terms of $\mathcal{L}_{\mathcal{S}}$ and $\text{clo}(\mathcal{S})$ the set of all closed terms of $\mathcal{L}_{\mathcal{S}}$.

Given $\varphi \in \text{For}(\mathcal{S})$ we denote by $\text{free}(\varphi)$ the set of all variables that occur free in φ and by $\text{free}(x, \varphi)$ the set of all terms free for the variable x in φ . The complexity of a formula (term) is defined as usual.

DEFINITION 5.1. A 4-structure over the first-order signature $\mathcal{S} = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$ is a pair $\mathfrak{A} = \langle A, (\cdot)^{\mathfrak{A}} \rangle$ such that A is a nonempty set and $(\cdot)^{\mathfrak{A}}$ is a map such that:

- If $P \in \mathcal{P}$ is an n -ary predicate symbol, then $P^{\mathfrak{A}} : A^n \rightarrow \mathbf{4}$.
- If $f \in \mathcal{F}$ is an n -ary function symbol, then $f^{\mathfrak{A}} : A^n \rightarrow A$.
- If $c \in \mathcal{C}$, then $c^{\mathfrak{A}} \in A$.

DEFINITION 5.2. Given a nonempty set X , an *assignment* on X is a map $s : \mathcal{V} \rightarrow X$. If $\mathfrak{A}$ is a 4-structure over the signature $\mathcal{S}$ then an *assignment on $\mathfrak{A}$* is an assignment on the domain A of the 4-structure $\mathfrak{A}$. We denote by $S(\mathfrak{A})$ the set of all assignment on $\mathfrak{A}$, i.e. $S(\mathfrak{A}) = A^{\mathcal{V}}$.

Given $s \in S(\mathfrak{A})$ and $t \in \text{Ter}(\mathcal{S})$, the *value of the term t in $\mathfrak{A}$ by the assignment s* (denoted $t^{\mathfrak{A}}[s]$) is defined inductively as follows:

- If $t \in \mathcal{V}$, then $t^{\mathfrak{A}}[s] = s(t)$.
- If $t \in \mathcal{C}$, then $t^{\mathfrak{A}}[s] = t^{\mathfrak{A}}$.
- If t is $f(t_1, \dots, t_n)$, where $f \in \mathcal{F}$ and $t_i \in \text{Ter}(\mathcal{S})$, then $t^{\mathfrak{A}}[s] = f^{\mathfrak{A}}(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s])$.

REMARK 5.3. Given $s \in S(\mathfrak{A})$, $x \in \mathcal{V}$ and $a \in A$, we denote by s_x^a the assignment over $\mathfrak{A}$ that satisfies

$$s_x^a(y) = \begin{cases} s(y) & \text{if } y \neq x \\ a & \text{if } y = x \end{cases}$$

DEFINITION 5.4. A $\text{QBD}^{\odot}$ -structure over $\mathcal{S}$ is a pair $\langle \mathfrak{A}, \|\cdot\|_{\text{QBD}^{\odot}}^{\mathfrak{A}} \rangle$ such that $\mathfrak{A}$ is a 4-structure over $\mathcal{S}$ and $\|\cdot\|_{\text{QBD}^{\odot}}^{\mathfrak{A}} : \text{For}(\mathcal{S}) \rightarrow \mathbf{4}$ is map defined inductively as follows, for every $s \in S(\mathfrak{A})$:

1. $\|P(t_1, \dots, t_n)\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) = P^{\mathfrak{A}}(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s])$, where $P \in \mathcal{P}$ is an n -ary predicate symbol;
2. $\|\# \varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) = \#\|\varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s)$, where $\# \in \{\odot, \neg\}$;
3. $\|\varphi \# \psi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) = \|\varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) \# \|\psi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s)$, where $\# \in \{\vee, \wedge, \rightarrow\}$,
4. $\|\forall x \varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) = \inf(\{\|\varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s_x^a) : a \in A\})$;
5. $\|\exists x \varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s) = \sup(\{\|\varphi\|_{\text{QBD}^{\odot}}^{\mathfrak{A}}(s_x^a) : a \in A\})$.

REMARK 5.5. Given that $\mathbf{4}$ is a finite lattice, it is complete and it satisfies the following, for every $\{a_i : i \in I\} \subseteq \mathbf{4}$: $\neg \inf_{i \in I} a_i = \sup_{i \in I} \neg a_i$ and $\neg \sup_{i \in I} a_i = \inf_{i \in I} \neg a_i$. From this, $\neg \inf_{i \in I} \neg a_i = \sup_{i \in I} a_i$.

DEFINITION 5.6. Let $\mathfrak{A}$ be a $\text{QBD}^\odot$ -structure over $\mathcal{S}$ and $\varphi \in \text{For}(\mathcal{S})$. We say that $s \in S(\mathfrak{A})$ satisfies φ in the $\text{QBD}^\odot$ -structure $\mathfrak{A}$ if $\|\varphi\|_{\text{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\} = \mathcal{D}_{\text{BD}^\odot}$. We say that $\mathfrak{A}$ is a $\text{QBD}^\odot$ -model of φ if every assignment on $\mathfrak{A}$ satisfies φ . We denote by $\text{mod}_{\text{QBD}^\odot}(\varphi)$ the class of all $\text{QBD}^\odot$ -models of φ . Let $\emptyset \neq \Gamma \subseteq \text{For}(\mathcal{S})$, we say that $\mathfrak{A}$ is a $\text{QBD}^\odot$ -model for Γ if $\mathfrak{A} \in \bigcap_{\gamma \in \Gamma} \text{mod}_{\text{QBD}^\odot}(\gamma)$; we denote by $\text{mod}_{\text{QBD}^\odot}(\Gamma)$ the class of all models of Γ , i.e. $\text{mod}_{\text{QBD}^\odot}(\Gamma) = \bigcap_{\gamma \in \Gamma} \text{mod}_{\text{QBD}^\odot}(\gamma)$.

DEFINITION 5.7. The logic $\text{QBD}^\odot$ (over $\mathcal{S}$) is the pair $\langle \mathcal{L}_\mathcal{S}, \overline{\overline{\quad}}_{\text{QBD}^\odot} \rangle$ where the consequence operator is defined as follows: $\Delta \overline{\overline{\quad}}_{\text{QBD}^\odot} \varphi$ iff $\text{mod}_{\text{QBD}^\odot}(\Delta) \subseteq \text{mod}_{\text{QBD}^\odot}(\varphi)$. In this case, we say that φ is a $\text{QBD}^\odot$ -consequence of Δ and, if $\Delta = \emptyset$, we say that φ is $\text{QBD}^\odot$ -valid.

THEOREM 5.8. Let $\mathfrak{A}$ be a $\text{QBD}^\odot$ -structure and let $s \in S(\mathfrak{A})$. Then: $\|\exists x \varphi\|_{\text{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$ implies that $\|\varphi\|_{\text{QBD}^\odot}^\mathfrak{A}(s_x^a)$ for some $a \in A$.

PROOF. It is consequence of Definition 5.4(5) and Remark 5.5. ■

PROPOSITION 5.9. The quantifier $\exists$ is definable from $\forall$ and $\neg$ in $\text{QBD}^\odot$ as $\exists x \varphi := \neg \forall x \neg \varphi$.

PROOF. It is consequence of Definition 5.4(5) and Remark 5.5. ■

REMARK 5.10. The formulas $\exists x \varphi$ and $\sim \forall x \sim \varphi$ are not strongly equivalent in $\text{QBD}^\odot$, that is: despite being equivalent, their denotations are not necessarily identical in any structure, hence they are not intersubstitutable in every formula. To see this, consider the next example: let $\mathcal{S} = \langle \{P\}, \emptyset, \emptyset \rangle$ be the first-order signature where P is a unary predicate symbol, and consider the $\text{QBD}^\odot$ -structure $\mathfrak{A}$ where its domain is $A = \{a, b\}$ and such that $P^\mathfrak{A}(a) = 0$ and $P^\mathfrak{A}(b) = \mathbf{b}$. Then, $\|\exists x P(x)\|_{\text{QBD}^\odot}^\mathfrak{A} = \sup\{0, \mathbf{b}\} = \mathbf{b}$, but, on the other hand, $\|\sim \forall x \sim P(x)\|_{\text{QBD}^\odot}^\mathfrak{A} = \sim (\inf\{\sim 0, \sim \mathbf{b}\}) = \sim (\inf\{1, 0\}) = \sim 0 = 1$. From this, $\mathfrak{A}$ is a $\text{QBD}^\odot$ -model of $\odot \sim \forall x \sim P(x)$, but not of $\odot \exists x P(x)$. Hence $\odot \exists x P(x)$ and $\odot \sim \forall x \sim P(x)$ are not equivalent in $\text{QBD}^\odot$, even though $\exists x P(x)$ and $\sim \forall x \sim P(x)$ are.

6. The Deductive System $\text{HQBD}^\odot$

In this section, we shall present a syntactic version of $\text{QBD}^\odot$ in terms of the Hilbert-style calculus $\text{HQBD}^\odot$ which extends $\text{HBD}^\odot_T$. From now on, only

the universal quantifier $\forall$ will be considered in the language (see Remark 6.2 below). Hence, from now on $\text{For}(\mathcal{S})$ and $\text{Sent}(\mathcal{S})$ will refer to formulas and closed formulas over $\mathcal{S}$ and $\mathbb{P}_T^\rightarrow$ without occurrences of the existential quantifier $\exists$.

DEFINITION 6.1. Consider the first-order signature $\mathcal{S} = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$. The Hilbert-style calculus $\text{HQB}^\odot$ over the language $\text{For}(\mathcal{S})$ is the extension of $\text{HBD}^\odot_T$, expressed in the language $\text{For}(\mathcal{S})$, by adding the following:

Axioms

$$(\mathbf{AxA}) \quad \forall x\varphi \rightarrow \varphi[x/t], \quad t \in \text{free}(x, \varphi)$$

$$(\mathbf{AxB}) \quad \neg\forall x\varphi \rightarrow \neg\forall x\neg\neg\varphi$$

$$(\mathbf{AxC}) \quad \neg\varphi[x/t] \rightarrow \neg\forall x\varphi \quad t \in \text{free}(x, \varphi)$$

Rules of inference

$$(\forall\text{-In1}) \quad \frac{\alpha \rightarrow \beta}{\alpha \rightarrow \forall x\beta}, \quad x \notin \text{free}(\alpha)$$

$$(\forall\text{-In2}) \quad \frac{\beta \rightarrow \alpha}{\neg\forall x\neg\beta \rightarrow \alpha}, \quad x \notin \text{free}(\alpha)$$

REMARK 6.2. We chose to work only with the universal quantifier. Though (as it was established in the previous section) $\forall$ and $\exists$ are semantically interdefinable by means of the paraconsistent negation $\neg$, that is, $\exists x\psi \equiv \neg\forall x\neg\neg\psi$, properties such as $(\mathbf{AxB})$, $(\mathbf{AxC})$, or the rule of introduction of $\exists$, represented by $(\forall\text{-In2})$, need to be explicitly included: they cannot be proven from the other axioms. This is due to fact that the paraconsistent negation $\neg$ enjoys less “good” properties than the classical negation. On the other hand, the strong negation $\sim$ is classic, that is, it is explosive and complete; and so, one would be tempted to recover the axioms and rules that involve $\exists$ using $\sim$ instead of $\neg$ to define $\exists$. But we have the problem that formulas such as $\exists x\alpha$ and $\sim\forall x\sim\alpha$ are not (strongly) equivalent, as pointed out in Remark 5.10.

REMARK 6.3. Let $\mathcal{S}$ be a first-order signature. An *instance* of a propositional formula $\varphi(p_1, \dots, p_n)$ in $\text{For}(\mathbb{P}_T^\rightarrow)$ is a formula in $\text{For}(\mathcal{S})$ obtained from φ by performing a simultaneous substitution of each occurrence of the propositional variable p_i by the formula β_i in $\text{For}(\mathcal{S})$, for $1 \leq i \leq n$. Let us denote by $\varphi[p_1/\beta_1, \dots, p_n/\beta_n]$ such an instance.

THEOREM 6.4. *If $\varphi \in \text{For}(\mathcal{S})$ is an instance of a $\text{BD}^\odot_T$ -valid propositional formula, then $\vdash_{\text{HQB}^\odot} \varphi$.*

PROOF. It is a direct adaptation of [32, Proposition 2.1]. ■

6.1. Deduction Metatheorem for $\text{HQBD}^\odot$

DEFINITION 6.5. Let $\Gamma \subseteq \text{For}(\mathcal{S})$, $\varphi \in \Gamma$ and let $\psi_1 \dots \psi_n$ be a deduction of ψ_n from Γ in $\text{HQBD}^\odot$. We say that ψ_i *depends upon* φ in the deduction if it is verified one of the following conditions:

- (a) ψ_i is φ ;
- (b) ψ_i is a direct consequence by **(MP)**, **($\forall$ -In1)** or **($\forall$ -In2)** of some preceding formulas of the sequence, where at least one of these preceding formulas depends upon φ .

The proof of the following result can be done exactly as in the case of first-order classical logic, taking into account that $\text{HQBD}^\odot$ is an axiomatic extension of that logic. A detailed proof for the classical case can be found, for instance, in [32, Proposition 1.9].

PROPOSITION 6.6. (Deduction Metatheorem (MTD)) *If in some deduction showing that $\Gamma, \varphi \mid_{\text{HQBD}^\odot} \psi$ no application of (**$\forall$ -In1**) and (**$\forall$ -In2**) to a formula that depends upon φ has as its quantified variable a free variable of φ , then $\Gamma \mid_{\text{HQBD}^\odot} \varphi \rightarrow \psi$.*

COROLLARY 6.7. *Suppose that there is some deduction of $\Gamma, \varphi \mid_{\text{HQBD}^\odot} \psi$ involving no application of (**$\forall$ -In1**) and (**$\forall$ -In2**) in which the quantified variable is free in φ . Then, $\Gamma \mid_{\text{HQBD}^\odot} \varphi \rightarrow \psi$.*

COROLLARY 6.8. (Deduction Metatheorem for sentences) *If φ is a sentence and $\Gamma, \varphi \mid_{\text{HQBD}^\odot} \psi$, then $\Gamma \mid_{\text{HQBD}^\odot} \varphi \rightarrow \psi$.*

The following two propositions can be proven as in the case of first-order classical logic and first-order **LFIs** (see [9, Chapter 7]).

PROPOSITION 6.9. *The rule of generalization*

$$(\text{Gen}) \quad \frac{\alpha}{\forall x \alpha}$$

is derivable in $\text{HQBD}^\odot$.

PROPOSITION 6.10.

- (i) $\alpha \mid_{\text{HQBD}^\odot} \neg \forall x \neg \alpha[x/t]$
- (ii) $\sim \forall x \alpha \mid_{\text{HQBD}^\odot} \neg \forall x \neg \sim \alpha$
- (iii) $\neg \alpha \rightarrow \beta \mid_{\text{HQBD}^\odot} \neg \forall x \alpha \rightarrow \beta$, if $x \notin \text{free}(\beta)$

$$(iv) \frac{}{\text{HQBD}^\ominus} \forall x \sim \alpha \rightarrow \sim \forall x \neg \alpha$$

$$(v) \sim \neg \beta \frac{}{\text{HQBD}^\ominus} \sim \neg \forall x \beta$$

$$(vi) \frac{}{\text{HQBD}^\ominus} \sim \forall x \sim \alpha \rightarrow \neg \forall x \neg \alpha$$

$$(vii) \frac{}{\text{HQBD}^\ominus} (\alpha \rightarrow \neg \forall x \beta) \rightarrow \neg \forall x \neg (\alpha \rightarrow \neg \beta),$$

where $\alpha \in \text{Sent}(\mathcal{S})$ and $\text{free}(\beta) = \{x\}$

$$(viii) (\varphi \rightarrow \neg \phi) \rightarrow \psi \frac{}{\text{HQBD}^\ominus} (\varphi \rightarrow \neg \forall x \phi) \rightarrow \psi,$$

where $\varphi \in \text{Sent}(\mathcal{S})$, $\text{free}(\phi) = \{x\}$ and $x \notin \text{free}(\psi)$.

LEMMA 6.11. Let $\alpha = \varphi[p_1/\beta_1, \dots, p_n/\beta_n] \in \text{For}(\mathcal{S})$ be an instance of a formula $\varphi(p_1, \dots, p_n)$ in the propositional language $\text{For}(\mathbb{P}_T^\rightarrow)$. For every $s \in S(\mathfrak{A})$ we define the $\text{BD}^\ominus_T$ -morphism e_s as follows: $e_s(p_i) = \|\beta_i\|_{\text{QBD}^\ominus}^\mathfrak{A}(s)$ if $1 \leq i \leq n$, and $e_s(p_i) = 0$ if $i > n$. Then, $\|\alpha\|_{\text{QBD}^\ominus}^\mathfrak{A}(s) = e_s(\varphi)$.

PROOF. Using induction on the complexity of φ . ■

COROLLARY 6.12. If φ is an instance of an axiom of $\text{HBD}^\ominus_T$, then $\frac{}{\text{QBD}^\ominus} \varphi$.

LEMMA 6.13. Let $t(x_1, \dots, x_n)$ be a term and let $\alpha(x_1, \dots, x_n)$ be a formula in $\text{For}(\mathcal{S})$. If s and s' are assignment in $\mathfrak{A}$ such that $s(x_i) = s'(x_i)$ for all i ($1 \leq i \leq n$), then

$$(i) t^\mathfrak{A}[s] = t^\mathfrak{A}[s'],$$

$$(ii) \|\alpha\|_{\text{QBD}^\ominus}^\mathfrak{A}(s) = \|\alpha\|_{\text{QBD}^\ominus}^\mathfrak{A}(s'),$$

PROOF. See [33, Proposition 2.2]. ■

LEMMA 6.14. (Substitution Lemma) Let $t \in \text{free}(x, \varphi)$, s an assignment in $\mathfrak{A}$ and $a = t^\mathfrak{A}[s]$. Then, $\|\varphi[x/t]\|_{\text{QBD}^\ominus}^\mathfrak{A}(s) = \|\varphi\|_{\text{QBD}^\ominus}^\mathfrak{A}(s_x^a)$.

PROOF. The proof is standard, by induction on the complexity of φ . ■

LEMMA 6.15. Let $t \in \text{free}(x, \varphi)$, then

$$(a) \frac{}{\text{QBD}^\ominus} \forall x \varphi \rightarrow \varphi[x/t],$$

$$(b) \frac{}{\text{QBD}^\ominus} \neg \forall x \varphi \rightarrow \neg \forall x \neg \neg \varphi,$$

$$(c) \frac{}{\text{QBD}^\ominus} \neg \varphi[x/t] \rightarrow \neg \forall x \varphi.$$

PROOF. (a) Suppose that $\|\forall x \varphi\|_{\text{QBD}^\ominus}^\mathfrak{A}(s) = \inf_{a \in A} \|\varphi\|_{\text{QBD}^\ominus}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$. By definition of the order in 4, $\|\varphi\|_{\text{QBD}^\ominus}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$ for every a . In particular, $\|\varphi\|_{\text{QBD}^\ominus}^\mathfrak{A}(s_x^b) = \|\varphi[x/t]\|_{\text{QBD}^\ominus}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$ for $b = t^\mathfrak{A}[s]$, by Lemma 6.14.

(b) It is immediate from the fact that $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) = \|\neg\neg\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s)$ for every $s \in S(\mathfrak{A})$.

(c) Suppose that $\|\neg\varphi[x/t]\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{\mathbf{b}, 1\}$. Then, $\|\neg\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^b) \in \{\mathbf{b}, 1\}$ for $b = t^\mathfrak{A}[s]$, by Lemma 6.14. But then $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^b) \in \{\mathbf{b}, 0\}$ and so $\|\forall x\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) = \inf_{a \in A} \|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^a) \in \{\mathbf{b}, 0\}$, by definition of the order in 4. Hence $\|\neg\forall x\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{\mathbf{b}, 1\}$. ■

LEMMA 6.16.

$$(1) \varphi \rightarrow \psi, \varphi \mid_{\mathbf{QBD}^\odot} \psi$$

(2) If $x \notin \text{free}(\varphi)$, then

$$(2a) \varphi \rightarrow \psi \mid_{\mathbf{QBD}^\odot} \varphi \rightarrow \forall x\psi \quad \text{and} \quad (2b) \psi \rightarrow \varphi \mid_{\mathbf{QBD}^\odot} \neg\forall x\neg\psi \rightarrow \varphi$$

PROOF. (1) Analogous to the proof done for the propositional case.

(2a) Suppose that $\mathfrak{A} \in \text{mod}_{\mathbf{QBD}^\odot}(\varphi \rightarrow \psi)$. Let us pick $s \in S(\mathfrak{A})$, then $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$. Since $x \notin \text{free}(\varphi)$, for all $a \in A$, we have $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$ (by Lemma 6.13). Therefore, for all $a \in A$ we have that $\|\psi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$. Hence, $\|\forall x\psi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$ and then $\mathfrak{A} \in \text{mod}_{\mathbf{QBD}^\odot}(\varphi \rightarrow \forall x\psi)$.

(2b) Suppose that $\mathfrak{A} \in \text{mod}_{\mathbf{QBD}^\odot}(\psi \rightarrow \varphi)$. Let $s \in A^\mathcal{V}$ be an arbitrary assignment such that $\|\neg\forall x\neg\psi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$. Then, by Theorem 5.8 and Proposition 5.9, there is some $a \in A$ such that $\|\psi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$. Then, $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s_x^a) \in \{1, \mathbf{b}\}$. By Lemma 6.13, since $x \notin \text{free}(\varphi)$ we have $\|\varphi\|_{\mathbf{QBD}^\odot}^\mathfrak{A}(s) \in \{1, \mathbf{b}\}$. Then, $\mathfrak{A} \in \text{mod}_{\mathbf{QBD}^\odot}(\neg\forall x\neg\psi \rightarrow \varphi)$. ■

THEOREM 6.17. (Soundness) If $\Gamma \mid_{\mathbf{HQBD}^\odot} \varphi$ then $\Gamma \mid_{\mathbf{QBD}^\odot} \varphi$.

PROOF. By induction on the length of the derivation of φ from Γ in $\mathbf{HQBD}^\odot$, using Lemmas 6.15 and 6.16. ■

7. Henkin Sets of Formulas and Completeness

Recall that a *theory* in $\mathbf{HQBD}^\odot$ is any set of formulas in $\text{For}(\mathcal{S})$.

LEMMA 7.1. Let $\varphi \in \text{Sent}(\mathcal{S})$ and let $\Delta \subseteq \text{For}(\mathcal{S})$ be a φ -saturated set in $\mathbf{HQBD}^\odot$ (see Section 2). Then, either $\neg\varphi \wedge \odot\varphi \in \Delta$, or $\neg\odot\varphi \wedge \odot\varphi \in \Delta$.

PROOF. Necessarily, it holds that $\odot\varphi \in \Delta$. Otherwise, since Δ is a φ -saturated theory, we have that $\Delta, \odot\varphi \mid_{\mathbf{HQBD}^\odot} \varphi$ and, by virtue of Proposition 4.5(i), (DMT) (Corollary 6.8) and MP we conclude that $\Delta \mid_{\mathbf{HQBD}^\odot} \varphi$, which is a contradiction since Δ is φ -saturated.

On the other hand, suppose that $\neg\varphi \notin \Delta$ and $\neg\odot\varphi \notin \Delta$. Since Δ is φ -saturated we have

$$\Delta, \neg\varphi \mid_{\text{HQBD}^\odot} \varphi \text{ and } \Delta, \neg\odot\varphi \mid_{\text{HQBD}^\odot} \varphi.$$

Using Corollary 6.8 once again and axiom **Pos5**, we obtain that $\Delta \mid_{\text{HQBD}^\odot} (\neg\varphi \rightarrow \varphi) \wedge (\neg\odot\varphi \rightarrow \varphi)$ and, by Proposition 4.5(ii), it follows that $\Delta \mid_{\text{HQBD}^\odot} \varphi$, which is a contradiction.

Finally, both conditions cannot hold simultaneously since, by Proposition 4.5 items (iii), (vi) and (vii), Δ would be a trivial theory, which is a contradiction. ■

DEFINITION 7.2. A theory Γ is said to be *maximal* in $\text{HQBD}^\odot$ if it is $\perp$ -saturated for some formula $\perp$ such that $\perp \mid_{\text{HQBD}^\odot} \beta$ for every β .

So, Γ is maximal iff Γ is not trivial, but $\Gamma \cup \{\beta\}$ is trivial for any $\beta \notin \Gamma$.

PROPOSITION 7.3. Let Γ be a theory in $\text{HQBD}^\odot$.

- (1) If Γ is maximal then it is φ -saturated in $\text{HQBD}^\odot$, for every formula φ such that $\varphi \notin \Gamma$.
- (2) If Γ is φ -saturated in $\text{HQBD}^\odot$ for some formula φ then it is maximal in $\text{HQBD}^\odot$.
- (3) Γ is φ -saturated in $\text{HQBD}^\odot$ for some formula φ if and only if it is maximal in $\text{HQBD}^\odot$.

PROOF. (1) Suppose that $\varphi \notin \Gamma$. Since Γ is deductively closed, $\Gamma \nmid_{\text{HQBD}^\odot} \varphi$. Now, let $\psi \notin \Gamma$. Then, $\Gamma, \psi \mid_{\text{HQBD}^\odot} \perp$ and so $\Gamma, \psi \mid_{\text{HQBD}^\odot} \varphi$, since $\perp \mid_{\text{HQBD}^\odot} \varphi$.

(2) Assume that Γ is φ -saturated in $\text{HQBD}^\odot$. Observe that $\Gamma \nmid_{\text{HQBD}^\odot} \perp$: otherwise $\Gamma \mid_{\text{HQBD}^\odot} \varphi$, a contradiction. By Theorem 2.1, there exists a $\perp$ -saturated set Δ (that is, a maximal theory) which contains Γ . Observe that $\varphi \notin \Delta$: otherwise, since either $\neg\varphi \wedge \odot\varphi \in \Gamma$ or $\neg\odot\varphi \wedge \odot\varphi \in \Gamma$, by Lemma 7.1, it would follow that either $\perp_\varphi := \varphi \wedge \neg\varphi \wedge \odot\varphi \in \Delta$ or $\perp'_\varphi := \varphi \wedge \neg\odot\varphi \wedge \odot\varphi \in \Delta$. But both formulas $\perp_\varphi$ and $\perp'_\varphi$ are bottom in $\text{HQBD}^\odot$ by axiom ($\odot$ -1) and Proposition 4.5(iii), and so Δ would be trivial, a contradiction. Hence, $\varphi \notin \Delta$.

Suppose now that there exists $\beta \in \Delta$ such that $\beta \notin \Gamma$. Then, $\Gamma, \beta \mid_{\text{HQBD}^\odot} \varphi$. But then, $\Delta \mid_{\text{HQBD}^\odot} \varphi$ and so $\varphi \in \Delta$, a contradiction. This shows that $\Delta = \Gamma$ and so Γ is maximal in $\text{HQBD}^\odot$.

(3) It follows by (2) and by Definition 7.2. ■

Because of the last result, from now on the terms “maximal theory in $\text{HQBD}^{\odot}$ ” and “ φ -saturated theory in $\text{HQBD}^{\odot}$ for some formula φ ” will be used indistinctly. Since maximal theories are deductively closed, we can prove without difficulty the next result.

LEMMA 7.4. *If $\Delta \subseteq \text{For}(\mathcal{S})$ is maximal in $\text{HQBD}^{\odot}$, then*

- (a) *If $\{\alpha \rightarrow \beta, \alpha\} \subseteq \Delta$, then $\beta \in \Delta$.*
- (b) *$\alpha \wedge \beta \in \Delta$ iff $\{\alpha, \beta\} \subseteq \Delta$.*
- (c) *$\neg(\alpha \wedge \beta) \in \Delta$ iff $\neg\alpha \in \Delta$ or $\neg\beta \in \Delta$.*
- (d) *If $\forall x \alpha \in \Delta$, then $\alpha[x/t] \in \Delta$ for every $t \in \text{clo}(\mathcal{S})$.*
- (e) *If $\neg\alpha[x/t] \in \Delta$ for some $t \in \text{clo}(\mathcal{S})$, then $\neg\forall x \alpha \in \Delta$.*

The next result will be needed for defining canonical models.

THEOREM 7.5. *Let $\Delta \subseteq \text{For}(\mathcal{S})$ be maximal in $\text{HQBD}^{\odot}$. Then, for all $\alpha \in \text{Sent}(\mathcal{S})$ it holds one, and only one, of the following conditions:*

- (1) $\alpha \wedge \odot \alpha \in \Delta$
- (2) $\alpha \wedge \neg\alpha \in \Delta$
- (3) $\odot \alpha \wedge \neg\odot \alpha \in \Delta$
- (4) $\neg\alpha \wedge \odot \alpha \in \Delta$

PROOF. Let $\alpha \in \text{For}(\mathcal{S})$ be an arbitrary formula. Then, $\alpha \in \Delta$ or $\alpha \notin \Delta$.

Case 1 If $\alpha \in \Delta$, by Proposition 4.5(iii) and the maximality of Δ , necessarily $\odot \alpha \wedge \neg\odot \alpha \notin \Delta$. Then, by Lemma 7.4(b), one of the formulas $\odot \alpha$, $\neg\odot \alpha$ does not belong to Δ .

Case 1.1 If $\neg\odot \alpha \notin \Delta$, by the maximality of Δ we have that $\Delta, \neg\odot \alpha \mid_{\text{HQBD}^{\odot}} \odot \alpha$ and, by Proposition 4.5(iv) it holds that $\Delta \mid_{\text{HQBD}^{\odot}} \odot \alpha$. Then, $\alpha \wedge \odot \alpha \in \Delta$.

Case 1.2 If $\odot \alpha \notin \Delta$, again, by the maximality of Δ it holds that $\Delta, \odot \alpha \mid_{\text{HQBD}^{\odot}} \neg\alpha$ and, by Proposition 4.5(v), $\Delta \mid_{\text{HQBD}^{\odot}} \neg\alpha$. Since Δ is closed, $\neg\alpha \in \Delta$. Then, $\alpha \wedge \neg\alpha \in \Delta$.

Case 2 $\alpha \notin \Delta$. Then Δ is α -saturated. Therefore, by Lemma 7.1, either $\odot \alpha \wedge \neg\odot \alpha \in \Delta$ or $\neg\alpha \wedge \odot \alpha \in \Delta$.

Finally, let us note that conditions (1)–(4) cannot hold simultaneously. Moreover, they are mutually exclusive, as a consequence of Proposition 4.5(iii) and axiom $(\odot - 1)$. ■

DEFINITION 7.6. Let Δ be an arbitrary set of formulas in the language $\text{For}(\mathcal{S})$, and let C be a nonempty set of constant symbols in the signature $\mathcal{S}$. We say that Δ has witnesses in C (or that it is a Henkin set) for $\text{HQBD}^{\odot}$ if it holds: for every sentence of the form $\neg\forall x \varphi$, there is a constant symbol $c \in C$ such that, if $\Delta \mid_{\text{HBD}^{\odot}} \neg\forall x \varphi$, then $\Delta \mid_{\text{HBD}^{\odot}} \neg\varphi[x/c]$.

THEOREM 7.7. (Theorem of Constants) *Let Δ be an arbitrary set of formulas in the language $\text{For}(\mathcal{S})$, and let $\frac{C}{\text{HQBD}^\odot}$ be the consequence relation of $\text{HQBD}^\odot$ on the signature $\mathcal{S}_C$, which is obtained from $\mathcal{S}$ by adding the new constant symbols of C . Then, for every $\varphi \in \text{For}(\mathcal{S})$,*

$$\Delta \mid_{\text{HBD}^\odot} \varphi \text{ iff } \Delta \mid_{\text{HQBD}^\odot}^C \varphi.$$

That is, $\text{HQBD}^\odot$ over $\mathcal{S}_C$ is a conservative extension of $\text{HQBD}^\odot$ over $\mathcal{S}$.

PROOF. Analogous to the proof of [9, Theorem 7.5.2]. ■

THEOREM 7.8. *Let $\Delta \subseteq \text{For}(\mathcal{S})$. Then, there exists $\Delta^W \subseteq \text{For}(\mathcal{S})$ with witnesses in the set C for $\text{HQBD}^\odot$ such that $\Delta \subseteq \Delta^W$ and*

$$\Delta \mid_{\text{HQBD}^\odot} \varphi \text{ iff } \Delta^W \mid_{\text{HQBD}^\odot}^C \varphi.$$

Besides, any extension of Δ^W by sentences in the signature $\mathcal{S}_C$ is a set with witnesses in C .

PROOF. Let Δ be a set of formulas in $\text{For}(\mathcal{S})$ and let C be a set of new constant symbols with the same cardinality (say, κ) than $\text{For}(\mathcal{S})$. Let $\mathcal{S}_C$ be the first-order signature obtained from $\mathcal{S}$ by adding the set C of new constant symbols. Let $\{\psi_\xi(x_{i_\xi})\}_{\xi < \kappa}$ be a sequence of all formulas of $\text{For}(\mathcal{S}_C)$ with (at most) one free variable (if ψ_ξ is closed we take $x_{i_\xi} = x_1$). We choose a sequence $\{a_\xi\}_{\xi < \kappa}$ of elements in C in such a way that:

- a_ζ does not occur in any of the formulas $\psi_\xi(x_{i_\xi})$, for $\xi < \zeta$; and
- each a_ζ is different from a_ξ , for $\xi < \zeta$.

Now, consider for $\xi < \kappa$ the following sentences:

$$(S_\xi) \quad \neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \psi_\xi[x_{i_\xi}/a_\xi].$$

Let $\Delta^W := \Delta \cup \{(S_\xi) : \xi < \kappa\}$. By construction, we have that $\Delta \subseteq \Delta^W \subseteq \text{For}(\mathcal{S}_C)$ and Δ^W has witnesses in C (see Definition 7.6). Observe that, in Δ^W , each a_ξ occurs at most in the sentence (S_ξ) . Let us see that Δ^W is a conservative extension of Δ . Indeed, let $\varphi \in \text{For}(\mathcal{S})$ and suppose that $\Delta^W \mid_{\text{HQBD}^\odot}^C \varphi$. Since $\text{HQBD}^\odot$ is finitary, there exists a finite set $\Delta_0 \subseteq \Delta^W$ such that $\Delta_0 \mid_{\text{HQBD}^\odot}^C \varphi$. Besides, Δ_0 has a finite number of formulas (S_ξ) .

Suppose that the formula (S_ξ) is in Δ_0 and let $\Delta_1 := \Delta_0 \setminus \{(S_\xi)\}$. By Corollary 6.8, $\Delta_1 \mid_{\text{HQBD}^\odot}^C (\neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \psi[x_{i_\xi}/a_\xi]) \rightarrow \varphi$.

By a standard technique that can be found in the literature (see for instance [9, Theorem 7.5.2]), we can construct, from a given deduction in $\text{HQBD}^\odot$ of $(\neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \psi[x_{i_\xi}/a_\xi]) \rightarrow \varphi$ from Δ_1 , a deduction in $\text{HQBD}^\odot$

showing that

$$\Delta_1 \mid_{\text{HQBD}^\odot}^C (\neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \psi_\xi[x_{i_\xi}/y]) \rightarrow \varphi$$

where y is a variable symbol which does not occur in $(\neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \psi_n[x_{i_\xi}/a_\xi]) \rightarrow \varphi$.

From Proposition 6.10(viii), it follows that

$$\Delta_1 \mid_{\text{HQBD}^\odot}^C (\neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \forall y \psi_\xi[x_{i_\xi}/y]) \rightarrow \varphi.$$

By (AxC), we have $\mid_{\text{HQBD}^\odot}^C \neg \psi_{i_\xi}(x_{i_\xi}) \rightarrow \neg \forall y \psi_{i_\xi}[x_{i_\xi}/y]$, and by Proposition 6.10(iii), $\mid_{\text{HQBD}^\odot}^C \neg \forall x_{i_\xi} \psi_\xi \rightarrow \neg \forall y \psi_\xi[x_{i_\xi}/y]$. Therefore

$$\Delta_1 \mid_{\text{HQBD}^\odot}^C \varphi.$$

Repeating this procedure a finite number n of times (since, as noted above, Δ_0 has a finite number of formulas (S_ξ)) we conclude that

$$\Delta_n \mid_{\text{HQBD}^\odot}^C \varphi,$$

where $\Delta_n \subseteq \Delta$. Then, $\Delta \mid_{\text{HQBD}^\odot}^C \varphi$ and by Theorem 7.7, we have $\Delta \mid_{\text{HQBD}^\odot} \varphi$. Then, Δ^W is a conservative extension of Δ . ■

DEFINITION 7.9. A set of sentences Θ in the language $\text{For}(\mathcal{S})$ is said to be a *maximal set of sentences* (in short, an mss) in $\text{HQBD}^\odot$ if there is a φ -saturated set $\Delta \subseteq \text{For}(\mathcal{S})$ in $\text{HQBD}^\odot$ (for some φ) such that $\Theta = \Delta \cap \text{Sent}(\mathcal{S})$.

THEOREM 7.10. (Henkin's model existence) *Let Θ be an mss with witnesses in $\text{HQBD}^\odot$. Then, Θ has a $\text{QBD}^\odot$ -model.*

PROOF. We define the $\text{QBD}^\odot$ -structure $\mathfrak{A}$ on $\mathcal{S}$ as follows:

- the domain A is the set of all closed terms in $\mathcal{S}$, $\text{clo}(\mathcal{S})$;
- if c is a constant symbol of $\mathcal{S}$, $c^\mathfrak{A} := c$;
- if f is an n -ary function symbol of $\mathcal{S}$, $f^\mathfrak{A} : A^n \rightarrow A$ is given by $f^\mathfrak{A}(a_1, \dots, a_n) := f(a_1, \dots, a_n)$;
- if R is an n -ary predicate symbol of $\mathcal{S}$, $R^\mathfrak{A}$ is given as follows:

$$\begin{array}{ll} R^\mathfrak{A}(a_1, \dots, a_n) = 1 & \text{iff} \quad R(a_1, \dots, a_n) \wedge \odot R(a_1, \dots, a_n) \in \Theta \\ R^\mathfrak{A}(a_1, \dots, a_n) = \mathbf{b} & \text{iff} \quad R(a_1, \dots, a_n) \wedge \neg R(a_1, \dots, a_n) \in \Theta \\ R^\mathfrak{A}(a_1, \dots, a_n) = \mathbf{n} & \text{iff} \quad \odot R(a_1, \dots, a_n) \wedge \neg \odot R(a_1, \dots, a_n) \in \Theta \\ R^\mathfrak{A}(a_1, \dots, a_n) = 0 & \text{iff} \quad \neg R(a_1, \dots, a_n) \wedge \odot R(a_1, \dots, a_n) \in \Theta. \end{array}$$

Note that the predicates are well defined by virtue of Theorem 7.5. Then, for every sentence $\varphi \in \text{Sent}(\mathcal{S})$, the $\text{QBD}^\odot$ -structure $\mathfrak{A}$ satisfies the following property (P):

- | | | | |
|-----|--|-----|--|
| (1) | $\ \varphi\ _{\text{QBD}^\odot}^{\mathfrak{A}} = 1$ | iff | $\varphi \wedge \odot \varphi \in \Theta$ |
| (2) | $\ \varphi\ _{\text{QBD}^\odot}^{\mathfrak{A}} = \mathbf{b}$ | iff | $\varphi \wedge \neg \varphi \in \Theta$ |
| (3) | $\ \varphi\ _{\text{QBD}^\odot}^{\mathfrak{A}} = \mathbf{n}$ | iff | $\odot \varphi \wedge \neg \odot \varphi \in \Theta$ |
| (4) | $\ \varphi\ _{\text{QBD}^\odot}^{\mathfrak{A}} = 0$ | iff | $\neg \varphi \wedge \odot \varphi \in \Theta$. |

Indeed, we use induction on the complexity of the sentence φ . Observe that it is enough to prove only the “if” part of conditions (1)–(4), since the “only if” part follows from this, by Theorem 7.5.⁹

Base step. If φ is the atomic formula $R(a_1, \dots, a_n)$, (P) holds by the definition of the $\text{QBD}^\odot$ -structure and the fact that $\|R(a_1, \dots, a_n)\|_{\text{QBD}^\odot}^{\mathfrak{A}} = R^{\mathfrak{A}}(a_1, \dots, a_n)$.

Inductive step. Let φ a sentence of complexity $n \geq 1$. Then, we consider the following subcases:

- | | |
|-------------------------------|--|
| (a) φ is $\neg \psi$ | (c) φ is $\psi \# \chi$, where $\# \in \{\vee, \wedge, \rightarrow\}$ |
| (b) φ is $\odot \psi$ | (d) φ is $\forall x \psi$. |

We shall just analyze the subcase (d). The others are left to the patient reader.

Case d: φ is $\forall x \psi$.

(d.1) If $\forall x \psi \wedge \odot \forall x \psi \in \Theta$, then, by Lemma 7.4(b) and (d), we have $\psi[x/a] \in \Theta$ for all $a \in A$. By (I.H.), $\|\psi[x/a]\|_{\text{QBD}^\odot}^{\mathfrak{A}} \in \{1, \mathbf{b}\}$ for all $a \in A$. Suppose that there is $b \in A$ such that $\|\psi[x/b]\|_{\text{QBD}^\odot}^{\mathfrak{A}} = \mathbf{b}$. Then, by (I.H.), $\neg \psi[x/b] \in \Theta$ and then, by (AxC) and Lemma 7.4(iii), we get $\neg \forall x \psi \in \Theta$. This contradicts the fact that Θ is non trivial (recall Proposition 4.5(iii)), and therefore $\|\forall x \psi\|_{\text{QBD}^\odot}^{\mathfrak{A}} = 1$.

(d.2) If $\forall x \psi \wedge \neg \forall x \psi \in \Theta$ then, by Lemma 7.4(a) and (b), we know that

$$\mathbf{b} \leq \|\psi[x/a]\|_{\text{QBD}^\odot}^{\mathfrak{A}} \leq 1, \text{ for all } a \in A \text{ (I.H.)}.$$

On the other hand, from Lemma 7.4(b) it follows that $\neg \forall x \psi \in \Theta$ and then, there exists a witness $b \in A$ such that $\neg \psi[x/b] \in \Theta$. Now, by (I.H.), we have that $\|\psi[x/b]\|_{\text{QBD}^\odot}^{\mathfrak{A}} \in \{0, \mathbf{b}\}$. Then, $\|\psi[x/b]\|_{\text{QBD}^\odot}^{\mathfrak{A}} = \mathbf{b}$. By definition of the order in **4**, we have $\|\forall x \psi\|_{\text{QBD}^\odot}^{\mathfrak{A}} = \mathbf{b}$.

⁹Indeed, assuming that the “if” part of conditions (1)–(4) was already proved for a given φ , suppose that the left-hand side of (i) holds (for some $1 \leq i \leq 4$). Then the left-hand side of (j) does not hold, for $j \neq i$, and so the right-hand side of (j) does not hold, for $j \neq i$. By Theorem 7.5, the right-hand side of (i) must hold.

(d.3) Suppose that $\odot \forall x \psi \wedge \neg \odot \forall x \psi \in \Theta$. Since $\overline{\overline{\text{BD}^\odot}} (\odot \alpha \wedge \neg \odot \alpha) \rightarrow \sim \alpha$, as it is easy to check, then $\sim \forall x \psi \in \Theta$, by Lemma 7.4(a). Then, by Proposition 6.10(ii) and the fact that Θ is closed for sentences, we have $\neg \forall x \neg \sim \psi \in \Theta$, and therefore, there exists $b \in A$ such that $\neg \neg \sim \psi[x/b] \in \Theta$. Hence, by **(DNeg)**, $\sim \psi[x/b] \in \Theta$. Then, by Lemma 7.4(b) and (c), it holds one of the following cases: either $\neg \psi[x/b] \wedge \odot \psi[x/b] \in \Theta$ or $\odot \psi[x/b] \wedge \neg \odot \psi[x/b] \in \Theta$. By (I.H.), $\|\psi[x/b]\|_{\text{QBD}^\odot}^\mathfrak{A} \in \{\mathbf{n}, 0\}$. Now, suppose that $\|\psi[x/c]\|_{\text{QBD}^\odot}^\mathfrak{A} \in \{0, \mathbf{b}\}$ for some $c \in A$. By (I.H.), we have $\neg \psi[x/c] \in \Theta$, and then, by **(AxC)**, we have $\neg \forall x \psi \in \Theta$. Therefore Θ is trivial, taking into account that $(\odot \alpha \wedge \neg \odot \alpha \wedge \neg \alpha) \rightarrow \beta$ is provable in $\text{HBD}^\odot_T$, by Proposition 4.5. From this contradiction we conclude that $\|\psi[x/a]\|_{\text{QBD}^\odot}^\mathfrak{A} \in \{1, \mathbf{n}\}$, for all $a \in A$. Hence, $\|\psi[x/b]\|_{\text{QBD}^\odot}^\mathfrak{A} = \mathbf{n}$. Finally, by definition of the order in 4, $\|\forall x \psi\|_{\text{QBD}^\odot}^\mathfrak{A} = \mathbf{n}$.

(d.4) The proof of this case is analogous to case **(d.3)**.

Therefore, we have proved the property (P).

Let $\theta \in \Theta$. Since Θ is an mss it follows, by Theorem 7.5, that either $\theta \wedge \odot \theta \in \Theta$ or $\theta \wedge \neg \theta \in \Theta$. By the property (P), we conclude that $\|\theta\|_{\text{QBD}^\odot}^\mathfrak{A} \in \{1, \mathbf{b}\}$. Therefore, $\mathfrak{A}$ is a model of Θ . ■

THEOREM 7.11. (Completeness for sentences) *Let $\Delta \cup \{\varphi\} \subseteq \text{Sent}(\mathcal{S})$, then*

$$\Delta \overline{\overline{\text{QBD}^\odot}} \varphi \text{ implies that } \Delta \overline{\overline{\text{HQBD}^\odot}} \varphi.$$

PROOF. Suppose that $\Delta \not\overline{\overline{\text{HQBD}^\odot}} \varphi$. By Theorem 7.8, we can (conservatively) extend Δ to a set $\Delta^W \subseteq \text{For}(\mathcal{S}_C)$ with witnesses in $C \neq \emptyset$. Since $\Delta^W \not\overline{\overline{\text{HQBD}^\odot}}^C \varphi$, by Theorem 2.1 there exists a φ -saturated set $\Phi \subseteq \text{For}(\mathcal{S}_C)$ in $\text{HQBD}^\odot$ such that $\Delta^W \subseteq \Phi$. Hence, $\Theta = \Phi \cap \text{Sent}(\mathcal{S}_C)$ is an mss in $\text{HQBD}^\odot$ (recall Definition 7.9) which extends Δ . Moreover, Θ has witnesses in $\text{HQBD}^\odot$. Then, by Theorem 7.10, we know that there is a $\text{HQBD}^\odot$ -model $\mathfrak{A}_C$ of Θ . On the other hand, by Theorem 7.5 and the fact that $\varphi \notin \Phi$, either $\odot \varphi \wedge \neg \odot \varphi \in \Theta$ or $\neg \varphi \wedge \odot \varphi \in \Theta$. Let $\mathfrak{A}$ be the reduct of $\mathfrak{A}_C$ to $\mathcal{S}$. Hence, $\|\varphi\|_{\text{QBD}^\odot}^\mathfrak{A} = \|\varphi\|_{\text{QBD}^\odot}^{\mathfrak{A}_C} = \{\mathbf{n}, 0\}$. Since $\mathfrak{A}$ is a model of Δ , it is not the case that $\Delta \overline{\overline{\text{QBD}^\odot}} \varphi$. ■

REMARK 7.12. Observe that completeness was proved only for sentences, while soundness was stated for formulas in general (recall Theorem 6.17). However, a completeness theorem for formulas in general (i.e., for formulas possibly having free variables) can be obtained from Theorem 7.11 by observing the following: for any formula ψ let $(\forall)\psi$ be the universal closure of ψ , that is: if ψ is a sentence then $(\forall)\psi = \psi$, and if ψ has exactly the variables $x_1, \dots, x_n$ occurring free then $(\forall)\psi = (\forall x_1) \dots (\forall x_n) \psi$. If Γ is a set of formulas then $(\forall)\Gamma := \{(\forall)\psi : \psi \in \Gamma\}$. Thus, it is easy to see that,

for every set $\Gamma \cup \{\varphi\}$ of formulas: $\Gamma \vdash_{\text{QBD}^\odot} \varphi$ iff $(\forall)\Gamma \vdash_{\text{HQBD}^\odot} (\forall)\varphi$, and $\Gamma \vdash_{\text{QBD}^\odot} \varphi$ iff $(\forall)\Gamma \vdash_{\text{HQBD}^\odot} (\forall)\varphi$. From this, a general completeness result follows (see Corollary below).

COROLLARY 7.13. (Soundness and completeness theorem)
 $\Delta \vdash_{\text{QBD}^\odot} \varphi$ iff $\Delta \vdash_{\text{HQBD}^\odot} \varphi$.

PROOF. The “if” part is precisely Theorem 6.17. The proof for the “only if” part goes as follows:

$$\begin{array}{lcl} \Delta \vdash_{\text{QBD}^\odot} \varphi & \text{only if} & (\forall)\Delta \vdash_{\text{QBD}^\odot} (\forall)\varphi \quad \text{Remark 7.12} \\ & \text{only if} & (\forall)\Delta \vdash_{\text{HQBD}^\odot} (\forall)\varphi \quad \text{Theorem 7.11} \\ & \text{only if} & \Delta \vdash_{\text{HQBD}^\odot} \varphi \quad \text{Remark 7.12} \end{array}$$

■

THEOREM 7.14. (Compactness) *Suppose that $\Delta \vdash_{\text{QBD}^\odot} \varphi$. Then, there exists a finite set $\Delta_0 \subseteq \Delta$ such that $\Delta_0 \vdash_{\text{QBD}^\odot} \varphi$.*

PROOF. The proof is the standard one once we have established the soundness and completeness of a first-order semantics w.r.t. a compact proof system. Observe that our deductive system *à la Hilbert* $\text{HQBD}^\odot$ is compact by definition. ■

8. Concluding Remarks

This paper presents the four-valued logic $\text{BD}^\odot$, an expansion of Belnap-Dunn BD obtained by adding an unary connective $\odot$ which is an **LFI** consistency operator. The operator $\odot$ is the unique one such that, simultaneously: (1) $\text{BD}^\odot$ is maximal w.r.t. **LFI1**; (2) $\odot$ extends to BD the consistency operator $\circ$ of **LFI1**, still satisfying axiom **ciw**; and (3) $\odot$ allows to define an **LFU** undeterminedness operator. It is also proven that $\text{BD}^\odot$ coincides, up to signature, with an expansion of BD by adding a classical-like negation proposed by De and Omori, called BD2.

A quite natural Hilbert-style characterization of $\text{BD}^\odot$ is obtained by means of twist-structures semantics. In addition, a first-order version of $\text{BD}^\odot$, called $\text{QBD}^\odot$, is also proposed by means of a Hilbert-style calculus called $\text{HQBD}^\odot$. Finally, soundness and completeness theorems of $\text{HQBD}^\odot$, with respect to a semantics based on a suitable notion of four-valued Tarskian-like structures called **4**-structures, are presented.

Observe that in this paper the logic **BD** was expanded with an **LFI** consistency operator (while preserving the 3-valued **LFI1** consistency operator) as an **LFIU**. Moreover, it was shown that there is only one connective enjoying this (and another) features. It would be interesting to analyze the “symmetric” case. That is, departing from the 3-valued determinedness operator $\star$ defined over $\{0, \mathbf{n}, 1\}$ such that $\star(x) = 1$ iff $x \in \{0, 1\}$ and $\star(\mathbf{n}) = 0$, there are only two ways of extending this operator to **4** in order to get an expansion of **BD** by a classicality operator:

	$\circ$	$\circ_1$
1	1	1
b	0	n
n	0	0
0	1	1

Let $\mathbf{A}_{\mathbf{BD}^\circ}^\circ$ and $\mathbf{A}_{\mathbf{BD}^{\circ_1}}^{\circ_1}$ be the expansions of the algebra **DM4** by $\circ$ and $\circ_1$, respectively, and let $\mathbf{BD}^\circ$ and $\mathbf{BD}^{\circ_1}$ be the logics induced by the respective matrices with designated values as in **BD**. It was already proven that $\circ$ can be defined in $\mathbf{A}_{\mathbf{BD}^\circ}$, and it is easy to see that the operator $\circ$ can be defined in $\mathbf{A}_{\mathbf{BD}^\circ}^{\circ_1}$ as $\circ(x) = \circ_1(x) \wedge \circ_1 \circ_1(x)$. This means that the logic $\mathbf{BD}^\circ$ is contained in both $\mathbf{BD}^\circ$ and $\mathbf{BD}^{\circ_1}$. However, the expressive powers of $\mathbf{A}_{\mathbf{BD}^\circ}$ and $\mathbf{A}_{\mathbf{BD}^\circ}^{\circ_1}$ are incomparable. Indeed, since $\{0, \mathbf{b}, 1\}$ is a subalgebra of $\mathbf{A}_{\mathbf{BD}^\circ}$ then the latter cannot define the operator $\circ_1$. Analogously, given that $\{0, \mathbf{n}, 1\}$ is a subalgebra of $\mathbf{A}_{\mathbf{BD}^\circ}^{\circ_1}$ it follows that the latter cannot define the operator $\sim$. The previous considerations suggest that the study of the logics $\mathbf{BD}^\circ$ and $\mathbf{BD}^{\circ_1}$ could be an interesting task to be carried out in the future.

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